Revision of the Grade 8 course, and the entry diagnostic

Pythagoras, the triangle criteria and the circle — the four facts the whole year is built on.

LESSON 1–2 | 2 × 40 MIN | GEOMETRIYA 9, TAKRORLASH | IGX 3, 11 AUDIT

LESSON CLOCK

15' | 20' | 20' | 20' | 5'

Triangles	15 min
Right triangles	20 min
The circle	20 min
The diagnostic	20 min
Feedback	5 min

LEARNING OBJECTIVES

- Recall the three congruence criteria and use them in a short proof.
- Use Pythagoras' theorem and the trigonometric ratios in a right triangle.
- Recall the circle theorems of Grade 8 — the inscribed angle and the tangent.
- Identify personally weak areas before the similarity chapter begins.

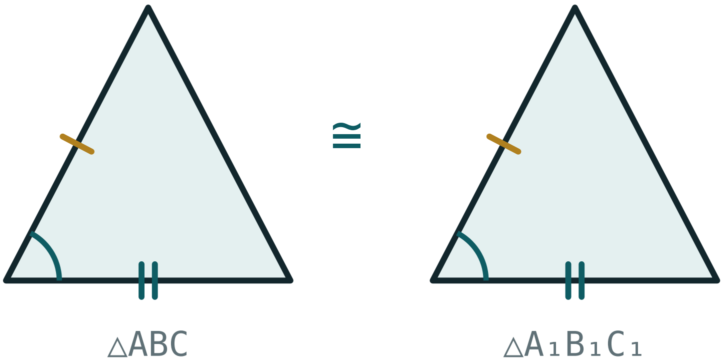
TEXTBOOK

National	pp. 3–8
Cambridge	Core & Extended, pp. 40–72, 220–240
Workbook	Diagnostic paper G0

01 Explanation

Triangles

CRITERION	NEEDS	SHORT NAME
first	two sides and the angle between them	SAS
second	a side and the two angles on it	ASA
third	three sides	SSS



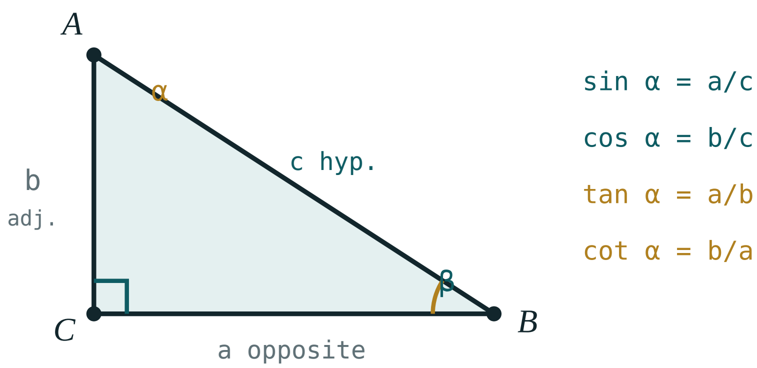
Two triangles are congruent when three correctly chosen elements agree.

The angles of a triangle add to $180°$; an exterior angle equals the sum of the two opposite interior angles; the angles opposite equal sides are equal.

> CONGRUENT NOW, SIMILAR NEXT
>
> This year the criteria are repeated with one word changed: instead of equal sides, **proportional** sides. Everything about the structure of a proof stays the same, which is why the revision matters more than usual.

Right triangles

$$a^2 + b^2 = c^2 \quad \sin A = \frac{a}{c} \quad \cos A = \frac{b}{c} \quad \tan A = \frac{a}{b}$$



Opposite, adjacent and hypotenuse — named from the angle you are working at.

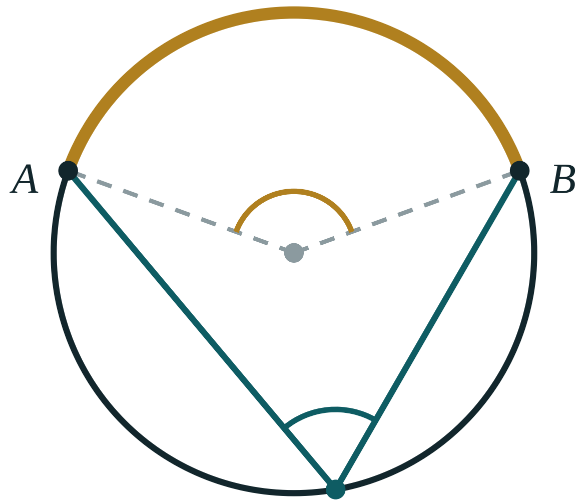
ANGLE	$30°$	$45°$	$60°$
sin	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$
cos	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$
tan	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$

THESE RATIOS ARE STILL ONLY FOR A RIGHT TRIANGLE

Quarter II removes that restriction with the sine and cosine rules. Until then, every use of sin, cos or tan requires a right angle somewhere in the figure — and if there is none, the first step is to draw an altitude and make one.

The circle

THEOREM	STATEMENT
inscribed angle	half the central angle on the same arc
angle in a semicircle	$90°$
angles on the same arc	equal
tangent and radius	perpendicular at the point of contact
two tangents from a point	equal in length



∠ACB = ½C∠AOB

The inscribed angle is half the central angle standing on the same arc.

These five return in Quarter III, when the circle is combined with regular polygons, and in Quarter IV, when chords and secants give proportional segments.

The diagnostic

Q	TASK	MARKS	TESTS
1	Find the third side of a right triangle with legs 9 and 12	3	Pythagoras
2	Find x if $\sin 30° = \frac{x}{10}$	3	ratios
3	Two angles of a triangle are $50°$ and $70°$: find the third	2	angle sum
4	A central angle is $80°$: find the inscribed angle on the same arc	3	the circle
5	Name the criterion that proves two triangles with three equal sides congruent	2	criteria
6	Find the area of a triangle with base 12 and height 7	2	area

WHERE GRADE 9 GEOMETRY IS GOING

Four chapters: similarity and transformations; trigonometry in any triangle; regular polygons and the circle; proportional segments. The first and the last are the same idea seen twice, and the middle two are what make geometry computational rather than only deductive.

02 Worked examples

EXAMPLE 1 **A right triangle has legs 9 and 12. Find the hypotenuse.**

1. $c^2 = 9^2 + 12^2$
2. $= 81 + 144 = 225$
3. $c = 15$
4. A 3–4–5 triangle, scaled by 3.

   Worth recognising.

ANSWER

15

EXAMPLE 2 **A central angle is $80°$. Find the inscribed angle on the same arc.**

1. The inscribed angle is half the central angle.
2. $$\frac{80°}{2}$$
3. $= 40°$
4. Every inscribed angle on that arc is $40°$.

   Not only one of them.

ANSWER

$40°$

EXAMPLE 3 **In a right triangle, $\sin A = 0.6$ and the hypotenuse is 20. Find the two legs.**

1. Opposite $= 20 \times 0.6 = 12$.
2. $\cos A = \sqrt{1 - 0.36} = 0.8$
3. Adjacent $= 20 \times 0.8 = 16$.
4. Check: $12^2 + 16^2 = 400 = 20^2$ ✓

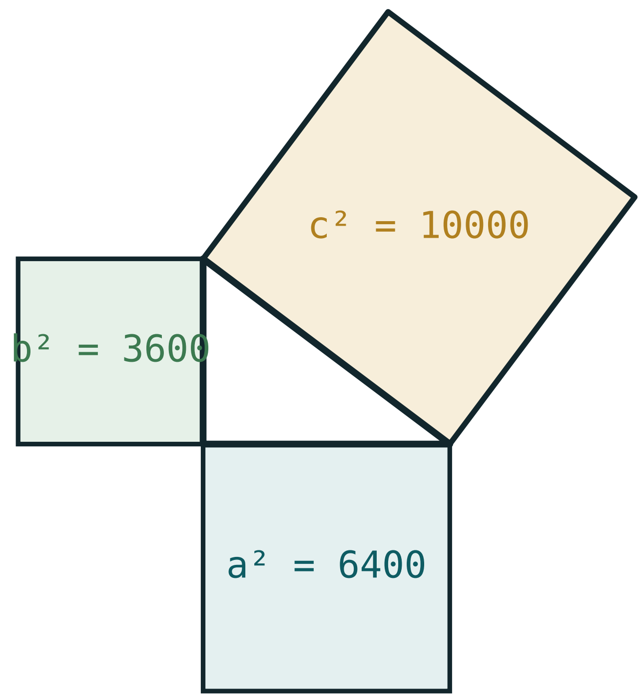
ANSWER

12 and 16

03 Interactive model

Draw one figure containing a right triangle, a circle and two congruent triangles, and ask the class to state every fact they can read from it; the diagnostic writes itself.

The three sides



a **80**	b **60**	c **100**	a^2 **6400**	b^2 **3600**	$a^2 + b^2$ **10000**	c^2 **10000**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Congruence	Tenglik	Равенство
Criterion	Alomat	Признак
Hypotenuse	Gipotenuza	Гипотенуза
Inscribed angle	Ichki chizilgan burchak	Вписанный угол
Central angle	Markaziy burchak	Центральный угол
Tangent	Urinma	Касательная
Median	Mediana	Медиана
Diagnostic test	Tashxis testi	Диагностический тест

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Three equal sides prove congruence by:

- SAS
- ASA
- SSS
- none

The angles of a triangle add to:

- $90°$
- $180°$
- $270°$
- $360°$

An inscribed angle is:

- equal to the central angle
- half of it
- twice it
- unrelated

The angle in a semicircle is:

- $45°$
- $60°$
- $90°$
- $180°$

A tangent meets the radius at:

- $30°$
- $45°$
- $60°$
- $90°$

$sin\ 60°$ equals:

- $\frac{1}{2}$
- $\frac{\sqrt{2}}{2}$
- $\frac{\sqrt{3}}{2}$
- 1

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Legs 9 and 12: hypotenuse

   ANSWER 15

2. Legs 5 and 12: hypotenuse

   ANSWER 13

3. Angles $50°$ and $70°$: the third

   ANSWER $60°$

4. Central angle $80°$: inscribed angle

   ANSWER $40°$

5. $\sin 30°$

   ANSWER $\frac{1}{2}$

6. $\tan 45°$

   ANSWER 1

7. Base 12, height 7: area

   ANSWER 42

Medium · the standard question

7 PROBLEMS

1. Hypotenuse 20, $\sin A = 0.6$: the legs

ANSWER 12 and 16

2. A ladder $5\ m$ long, foot $3\ m$ out: height reached

ANSWER $4\ m$

3. An inscribed angle is $35°$: the central angle

ANSWER $70°$

4. An equilateral triangle of side 6: its height

ANSWER $3\sqrt{3}$

5. Its area

ANSWER $9\sqrt{3}$

6. A square of diagonal 10: its side

ANSWER $5\sqrt{2}$

7. Two tangents from a point, one is 8: the other

ANSWER 8

Hard · stretch and reason

7 PROBLEMS

1. A triangle *13*, *14*, *15*: its area

   ANSWER 84

2. Its height on the side *14*

   ANSWER 12

3. A chord *16* in a circle of radius *10*: its distance from the centre

   ANSWER 6

4. An isosceles triangle with sides *13*, *13*, *10*: its area

   ANSWER 60

5. A rhombus with diagonals *16* and *12*: its side

   ANSWER 10

6. Its area

   ANSWER 96

7. A right triangle with legs a and b: the median to the hypotenuse

   ANSWER $\frac{\sqrt{a^2 + b^2}}{2}$

07 Homework

Homework — 5 tasks

Draw a labelled figure for every task; an unlabelled figure is worth no marks.

1. A right triangle has legs *7* and *24*. Find the hypotenuse.
2. An inscribed angle is *42°*. Find the central angle on the same arc.
3. Find the area of an equilateral triangle of side *8*.
4. A chord of length *12* lies in a circle of radius *10*. Find its distance from the centre.

5. State the three congruence criteria and draw a figure illustrating each.

Similarity of polygons

Same shape, different size — and the two conditions that say so precisely.

LESSON 3 | 1 × 40 MIN | GEOMETRIYA 9, §1 | IGX 11.3

LESSON CLOCK

10' | 12' | 12' | 6'

The definition	10 min
The coefficient	12 min
Both conditions are needed	12 min
Homework	6 min

LEARNING OBJECTIVES

- Define similar polygons by equal angles and proportional sides.
- Find and use the coefficient of similarity k.
- Show that both conditions are needed by a counter-example.
- Write a similarity statement with the vertices in matching order.

TEXTBOOK

National	pp. 9–13
Cambridge	Core & Extended, pp. 232–236
Workbook	Exercise 11.3

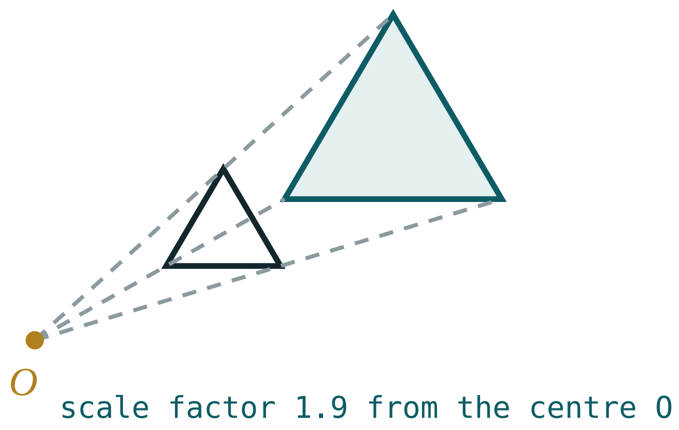
01 Explanation

The definition

Two polygons are **similar** if

1. their corresponding angles are equal, **and**
2. their corresponding sides are proportional.

$$ABCD \sim A_1B_1C_1D_1 \text{ means } \angle A = \angle A_1, \ldots \text{ and } \frac{\mathrm{AB}}{\mathrm{A_1B_1}} = \frac{\mathrm{BC}}{\mathrm{B_1C_1}} = \ldots = k$$



The same shape at two sizes — every length multiplied by k, every angle unchanged.

THE ORDER OF THE LETTERS IS PART OF THE STATEMENT

$ABCD \sim A_1B_1C_1D_1$ says that A corresponds to A_1, B to B_1, and so on. Writing $ABCD \sim B_1A_1C_1D_1$ claims something different, and usually something false.

The coefficient

The common ratio k is the **coefficient of similarity**. It scales every length in the figure — sides, diagonals, heights, medians, the perimeter.

QUANTITY	MULTIPLIED BY
any length	k
the perimeter	k
any angle	1
the area	k^2

$k > 1$ means an enlargement, $k < 1$ a reduction, and $k = 1$ congruence — so congruent figures are similar with coefficient 1.

AREAS GO AS K^2, AND THIS IS NOT OBVIOUS

Doubling every side multiplies the area by 4, not by 2. The full proof comes later in the chapter, but the fact is worth knowing from the first lesson — it is the single most-tested consequence of similarity.

Both conditions are needed

PAIR	ANGLES EQUAL?	SIDES PROPORTIONAL?	SIMILAR?
a square and a rectangle 2×4	yes	no	no
a square and a rhombus of the same side	no	yes	no
two squares	yes	yes	yes
two circles	—	—	always

The first two rows are the standard counter-examples, and they are worth drawing rather than merely reading: each condition on its own is genuinely insufficient.

TRIANGLES ARE THE EXCEPTION

For **triangles** either condition alone forces the other, which is why the next four lessons are about triangles specifically. For every other polygon, both must be checked.

02 Worked examples

EXAMPLE 1 **Two similar quadrilaterals have sides $3, 4, 5, 6$ and $9, 12, 15, x$. Find k and x.**

1. $k = \dfrac{9}{3} = 3$
2. Check: $\dfrac{12}{4} = 3$, $\dfrac{15}{5} = 3$ ✓
3. $x = 6 \times 3$
4. $= 18$

ANSWER

$k = 3, x = 18$

EXAMPLE 2 **The perimeter of a polygon is 20. A similar polygon has $k = 2.5$. Find its perimeter.**

1. Every side is multiplied by 2.5.
2. So is their sum.
3. 20×2.5
4. $= 50$

ANSWER

50

EXAMPLE 3 **Are a square of side 4 and a rectangle 4×8 similar?**

1. All angles are $90°$ — the first condition holds.
2. Sides: $\frac{4}{4} = 1$ but $\frac{4}{8} = 0.5$.
3. Not proportional.
4. No — the second condition fails.
   Both are needed.

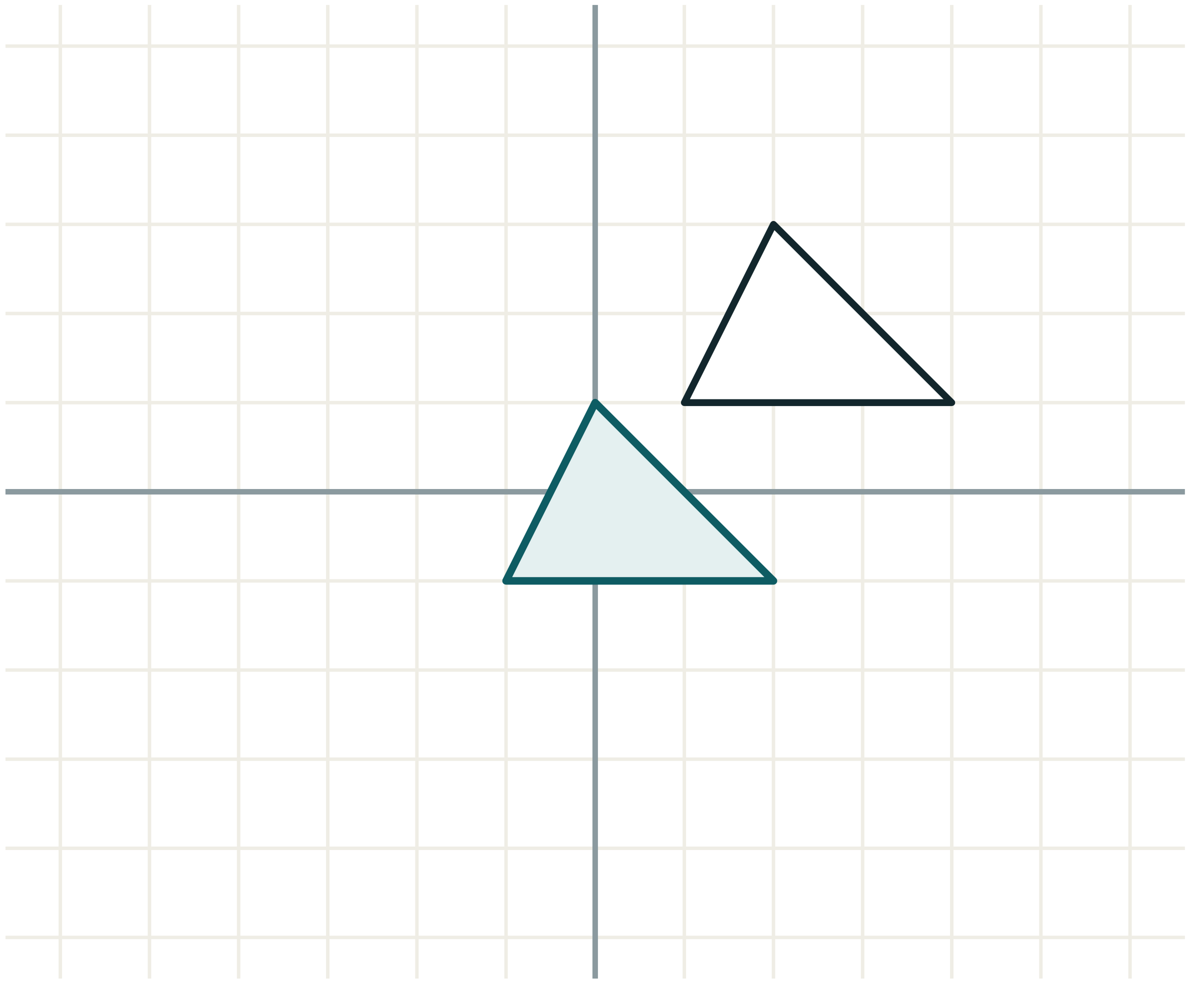
ANSWER

No

03 Interactive model

Hold up a photograph and an enlargement of it; ask what changed and what did not, and the two conditions are stated by the class before they are written down.

Enlarging a polygon



transformation **translation** | lengths **unchanged** | angles **unchanged** | orientation **kept**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Similar	O‘xshash	Подобный
Coefficient of similarity	O‘xshashlik koeffitsiyenti	Коэффициент подобия
Corresponding sides	Mos tomonlar	Соответственные стороны
Corresponding angles	Mos burchaklar	Соответственные углы
Proportional	Proporsional	Пропорциональные
Ratio	Nisbat	Отношение
Scale factor	Masshtab koeffitsiyenti	Масштабный коэффициент
Order of vertices	Uchlar tartibi	Порядок вершин

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Similar polygons have:

- equal sides
- proportional sides and equal angles
- equal areas
- the same perimeter

k is the ratio of:

- areas
- angles
- corresponding sides
- perimeters only

Under similarity the angles are:

- multiplied by k
- unchanged
- halved
- squared

The areas are in the ratio:

- k
- k^2
- k^3
- $2k$

$k = 1$ means the figures are:

- similar only
- congruent
- different
- not similar

A square and a rectangle 2×4 are:

- similar
- not similar
- congruent
- equal in area

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Sides 3 and 9: k

   ANSWER 3

2. Sides 10 and 4: k

   ANSWER 0.4

3. Perimeter 20, $k = 2.5$: new perimeter

   ANSWER 50

4. Angles under similarity

   ANSWER Unchanged

5. Areas ratio for $k = 3$

   ANSWER 9

6. Are two squares similar?

   ANSWER Always

7. Are two circles similar?

   ANSWER Always

Medium · the standard question

7 PROBLEMS

1. Sides $3,4,5,6$ and $9,12,15,x$: x

ANSWER 18

2. Are a square and a 4×8 rectangle similar?

ANSWER No

3. Are a square and a rhombus of the same side similar?

ANSWER No

4. Perimeters 12 and 30: k

ANSWER 2.5

5. Areas 16 and 144: k

ANSWER 3

6. A rectangle 3×5 and one 6×10

ANSWER Similar, $k = 2$

7. A rectangle 3×5 and one 6×9

ANSWER Not similar

Hard · stretch and reason

7 PROBLEMS

1. Two similar polygons have areas 50 and 98: k

   ANSWER $\frac{7}{5}$

2. Their perimeters are in the ratio

   ANSWER $7 : 5$

3. A photograph 10×15 enlarged to width 25: its height

   ANSWER 37.5

4. For which x are 2×6 and $3 \times x$ rectangles similar?

   ANSWER $x = 9$ (or $x = 1$)

5. A map scale $1 : 25000$: $4\ cm$ on the map

   ANSWER $1\ km$

6. Area on the map $2\ cm^2$: the real area

   ANSWER $0.125\ km^2$

7. Two similar figures: $k = \dfrac{2}{3}$. The ratio of their areas

   ANSWER $4 : 9$

07 Homework

Homework — 5 tasks

Write the similarity statement with the vertices in matching order every time.

1. Two similar pentagons have sides $2,3,4,5,6$ and $5, x, 10, y, 15$. Find k, x and y.
2. Two similar polygons have perimeters 18 and 45. Find k.
3. Their areas are in what ratio?
4. Draw a counter-example showing that equal angles alone do not give similarity.

5. A map has scale $1 : 50000$. Find the real distance for $6\ cm$ on the map.

Similar triangles and their properties

For triangles, one condition implies the other — which is what makes similarity a tool.

LESSON 4 | 1 × 40 MIN | GEOMETRIYA 9, §2 | IGX 11.2

LESSON CLOCK

10' | 12' | 12' | 6'

Why one condition is enough	10 min
Setting up the proportion	12 min
What else scales	12 min
Homework	6 min

LEARNING OBJECTIVES

- State what similarity means for triangles and why one condition suffices.
- Use the ratio of corresponding sides to find an unknown length.
- Know that heights, medians and bisectors scale by k, and areas by k^2.
- Set up a proportion with corresponding sides in matching positions.

TEXTBOOK

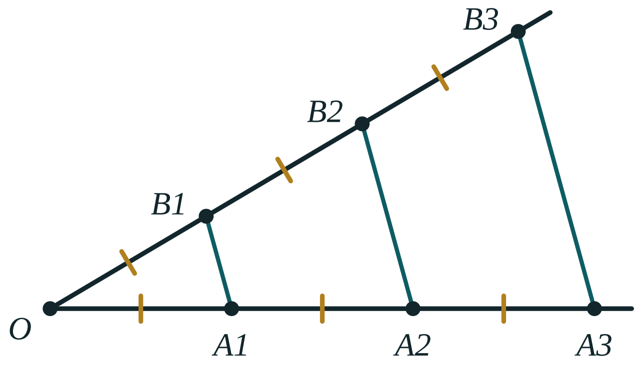
National	pp. 14–18
Cambridge	Core & Extended, pp. 226–231
Workbook	Exercise 11.2

01 Explanation

Why one condition is enough

For a general polygon, equal angles and proportional sides are independent conditions. For a **triangle** each forces the other.

IF	THEN
the angles are equal in pairs	the sides are automatically proportional
the sides are proportional	the angles are automatically equal



A line parallel to one side cuts the other two proportionally — the source of triangle similarity.

The reason is that a triangle is rigid: three sides determine it completely, and two angles determine its shape. No other polygon has that property, which is why the next three lessons can state criteria that need only two or three facts.

Setting up the proportion

If $\triangle ABC \sim \triangle A_1B_1C_1$ then

$$\frac{\mathrm{AB}}{\mathrm{A_1B_1}} = \frac{\mathrm{BC}}{\mathrm{B_1C_1}} = \frac{\mathrm{CA}}{\mathrm{C_1A_1}} = k$$

Corresponding sides are those **opposite equal angles**. Finding them correctly is the whole difficulty; the arithmetic afterwards is one cross-multiplication.

STEP	WHAT TO DO
1	mark the equal angles on both figures
2	write the similarity with the vertices in matching order
3	read the proportion straight off that statement
4	cross-multiply and solve

MISMATCHED SIDES GIVE A PLAUSIBLE BUT WRONG ANSWER

Pairing the longest side of one triangle with the shortest of the other still produces a number. Step 1 above is the guard, and it is why marking the angles on the diagram is not optional.

What else scales

ELEMENT	RATIO
corresponding sides	k
corresponding heights	k
corresponding medians	k
corresponding bisectors	k
perimeters	k
areas	k^2

The area result follows at once: $S = \frac{1}{2}ah$, and both a and h are multiplied by k, so S is multiplied by k^2.

EVERY *LENGTH* SCALES BY K — NO EXCEPTIONS

Including ones that are not sides: the radius of the inscribed circle, the distance from a vertex to the centroid, the length of any cevian. If it is measured in centimetres, it is multiplied by k; if in square centimetres, by k^2.

02 Worked examples

EXAMPLE 1 $\triangle ABC \sim \triangle A_1B_1C_1$ **with** $AB = 6$, $A_1B_1 = 9$, $BC = 8$. **Find** B_1C_1.

1. $k = \dfrac{9}{6} = 1.5$

   From the first pair.

2. $\dfrac{BC}{B_1C_1} = \dfrac{1}{1.5}$

3. $B_1C_1 = 8 \times 1.5$

4. $= 12$

ANSWER

12

EXAMPLE 2 **Two similar triangles have areas** 18 **and** 50**. Find the ratio of their perimeters.**

1. $k^2 = \dfrac{50}{18} = \dfrac{25}{9}$

2. $k = \dfrac{5}{3}$

   Take the square root.

3. Perimeters scale by k.

4. $5 : 3$

ANSWER

$5 : 3$

EXAMPLE 3

A triangle has a height of *12*. A similar triangle has sides $\frac{3}{4}$ as long. Find its height and the ratio of the areas.

1. $k = \frac{3}{4}$

2. Height $= 12 \times \frac{3}{4} = 9$.

   A length.

3. Areas: $k^2 = \frac{9}{16}$.

4. Height 9, areas $9 : 16$.

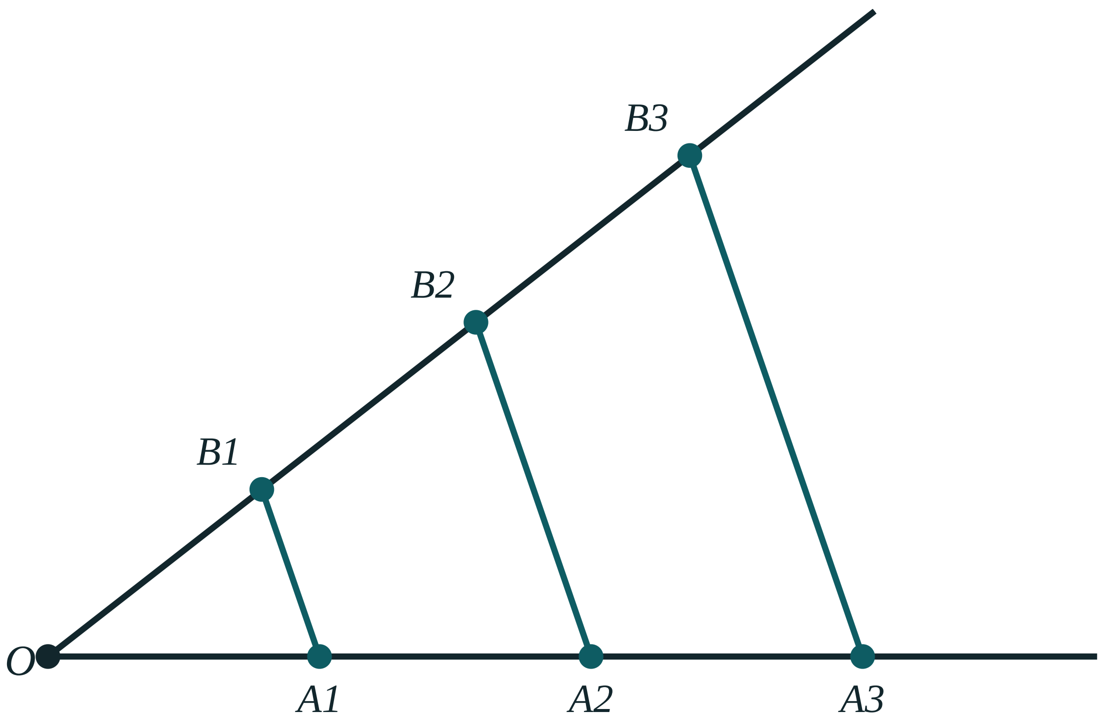
ANSWER

9; areas $9 : 16$

03 Interactive model

Draw two similar triangles in different orientations, one rotated; the class must match the angles before writing anything, and that is the whole skill.

Thales' theorem



OA_1 **88**	A_1A_2 **88**	OB_1 **88**	B_1B_2 **88**	$OA_1 : OA_2$ **1 : 2**	$OB_1 : OB_2$ **1 : 2**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Similar triangles	O‘xshash uchburchaklar	Подобные треугольники
Corresponding	Mos	Соответственный
Proportion	Proporsiya	Пропорция
Height	Balandlik	Высота
Median	Mediana	Медиана
Bisector	Bissektrisa	Биссектриса
Ratio of areas	Yuzalar nisbati	Отношение площадей
Cross-multiply	Krest usuli	Перекрёстное умножение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

For triangles, equal angles imply:

- nothing more
- proportional sides
- equal sides
- equal areas

Corresponding sides are those:

- of equal length
- opposite equal angles
- drawn first
- the longest

Corresponding medians are in the ratio:

- 1
- k
- k^2
- $2k$

Areas are in the ratio:

- k
- k^2
- k^3
- $2k$

Areas 18 and 50 give k =

- $\frac{25}{9}$
- $\frac{5}{3}$
- $\frac{3}{5}$
- $\frac{9}{25}$

The inscribed circle's radius scales by:

- 1
- k
- k^2
- it does not scale

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $AB = 6$, $A_1B_1 = 9$: k

   ANSWER 1.5

2. Same, $BC = 8$: B_1C_1

   ANSWER 12

3. $k = 2$: ratio of areas

   ANSWER 4

4. $k = 3$: ratio of perimeters

   ANSWER 3

5. Height 12, $k = \frac{3}{4}$: new height

   ANSWER 9

6. Median 10, $k = 0.6$: new median

   ANSWER 6

7. Areas $9 : 16$: k

   ANSWER $3 : 4$

Medium · the standard question

7 PROBLEMS

1. Areas 18 and 50: ratio of perimeters

   ANSWER $5 : 3$

2. Perimeters 15 and 25: ratio of areas

   ANSWER $9 : 25$

3. Sides $3, 5, 7$ and $9, x, y$

   ANSWER $x = 15, y = 21$

4. Areas 12 and 27: k

   ANSWER $\frac{3}{2}$

5. A triangle of area 20 scaled by $k = 2.5$

   ANSWER 125

6. Bisector 8, $k = 1.25$: new bisector

   ANSWER 10

7. Inradius 4, $k = 3$: new inradius

   ANSWER 12

Hard · stretch and reason

7 PROBLEMS

1. Two similar triangles: areas differ by 56, $k = 3$. The smaller area

   ANSWER 7

2. A triangle is cut by a midline: the ratio of the two areas

   ANSWER $1 : 3$

3. Similar triangles with perimeters 24 and 36, smaller area 32: the larger

   ANSWER 72

4. A model at $1 : 50$: the real area for $20\ cm^2$

   ANSWER $50\,000\ cm^2$

5. A triangle $6, 8, 10$ and a similar one of perimeter 36: its sides

   ANSWER $9, 12, 15$

6. Two similar triangles have heights 6 and 10: the ratio of areas

   ANSWER $9 : 25$

7. A triangle of area 60 is scaled so that its area is 15: find k

   ANSWER $\frac{1}{2}$

07 Homework

Homework — 5 tasks

Mark the equal angles on the figure before writing any proportion.

1. $\triangle ABC \sim \triangle DEF$ with $AB = 8$, $DE = 12$, $BC = 10$. Find EF.
2. Two similar triangles have areas 27 and 48. Find k.
3. A triangle of perimeter 30 is enlarged by $k = 1.4$. Find the new perimeter.
4. A triangle of area 45 is reduced by $k = \frac{1}{3}$. Find the new area.

5. Explain why the ratio of areas is k^2 and not k.

Home / Grade 9 / Geometry / Quarter 1

The first criterion of similarity — two angles

Two angles equal and the triangles are similar: the criterion used more than all the others together.

LESSON 5 | 1 × 40 MIN | GEOMETRIYA 9, §3 | IGX 11.2

LESSON CLOCK

10' | 14' | 10' | 6'

The criterion	10 min
Two configurations	14 min
Using it	10 min
Homework	6 min

LEARNING OBJECTIVES

- State and prove the AA criterion.
- Recognise the two standard configurations in which it appears.
- Use it to find an unknown length in a figure with parallel lines.
- Write a two-line proof of similarity from a diagram.

TEXTBOOK

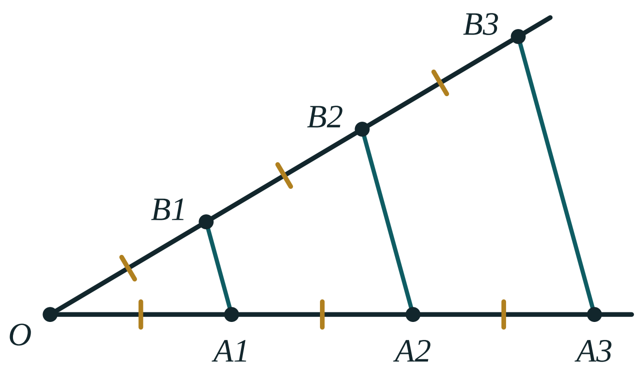
National	pp. 19–23
Cambridge	Core & Extended, pp. 226–231
Workbook	Exercise 11.2

01 Explanation

The criterion

> If two angles of one triangle equal two angles of another, the triangles are similar

The proof is one line: the third angles are equal too, because all three add to $180°$.
So all three angles agree, and for triangles that is enough.



Two angles equal — and the two triangles are the same shape at different sizes.

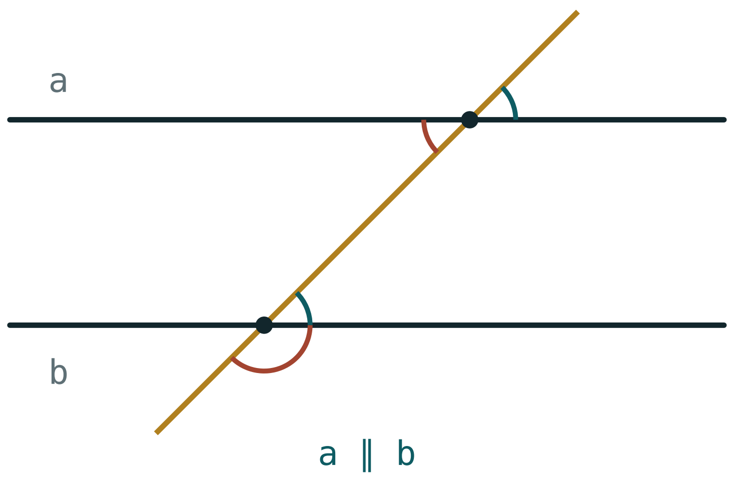
TWO ANGLES, AND NOTHING ABOUT SIDES AT ALL

This is what makes similarity useful. Angles are easy to find — from parallel lines, from vertical angles, from a shared angle — while sides usually have to be measured. Nine out of ten similarity problems are solved by this criterion.

Two configurations

Almost every diagram in this chapter is one of two pictures.

CONFIGURATION	THE EQUAL ANGLES COME FROM
the “*Λ*” — a line parallel to a side	the common angle, and corresponding angles
the “*X*” — two lines crossing between parallels	vertical angles, and alternate angles



Parallel lines cut by a transversal — the standard supply of equal angles.

In the Λ configuration, $DE \parallel BC$ in $\triangle ABC$ gives $\triangle ADE \sim \triangle ABC$ with $\angle A$ common. In the X configuration the equal angles are the vertical pair at the crossing point together with one pair of alternate angles.

> NAME THE VERTICES IN THE RIGHT ORDER IN THE X CASE
>
> The two triangles point in opposite directions, so the matching is not the obvious one. Mark the equal angles on the figure first, then read the order off the marks.

Using it

Example. In $\triangle ABC$, $DE \parallel BC$ with D on AB and E on AC. If $AD = 4$, $DB = 6$ and $DE = 5$, find BC.

STEP	WORKING
similar by AA	$\angle A$ common, $\angle ADE = \angle ABC$
k	$\frac{\mathrm{AB}}{\mathrm{AD}} = \frac{10}{4} = 2.5$
BC	$5 \times 2.5 = 12.5$

> AB IS 10, NOT 6
>
> The proportion involves the **whole** side $AB = AD + DB$, not the part beyond D. Using DB in place of AB is the classic error of this configuration.

02 Worked examples

EXAMPLE 1 **In $\triangle ABC$, $DE \parallel BC$, $AD = 4$, $DB = 6$, $DE = 5$. Find BC.**

1. $\angle A$ is common; $\angle ADE = \angle ABC$ (corresponding).
   AA.

2. $AB = 4 + 6 = 10$
   The whole side.

3. $k = \dfrac{10}{4} = 2.5$

4. $BC = 5 \times 2.5 = 12.5$

ANSWER

12.5

EXAMPLE 2 **Two triangles have angles $50°$, $60°$ and $60°$, $70°$. Are they similar?**

1. First triangle: third angle $70°$.

2. Second: third angle $50°$.

3. Both have $\{50°, 60°, 70°\}$.

4. Yes — similar by AA.

ANSWER

Yes

EXAMPLE 3 **A *1.8 m* person casts a *2.4 m* shadow; a tower casts a *20 m* shadow. Find the tower's height.**

1. Both triangles are right-angled with the same sun angle.
   AA.

2. $$\frac{h}{20} = \frac{1.8}{2.4}$$

3. $h = 20 \times 0.75$

4. $= 15\ m$

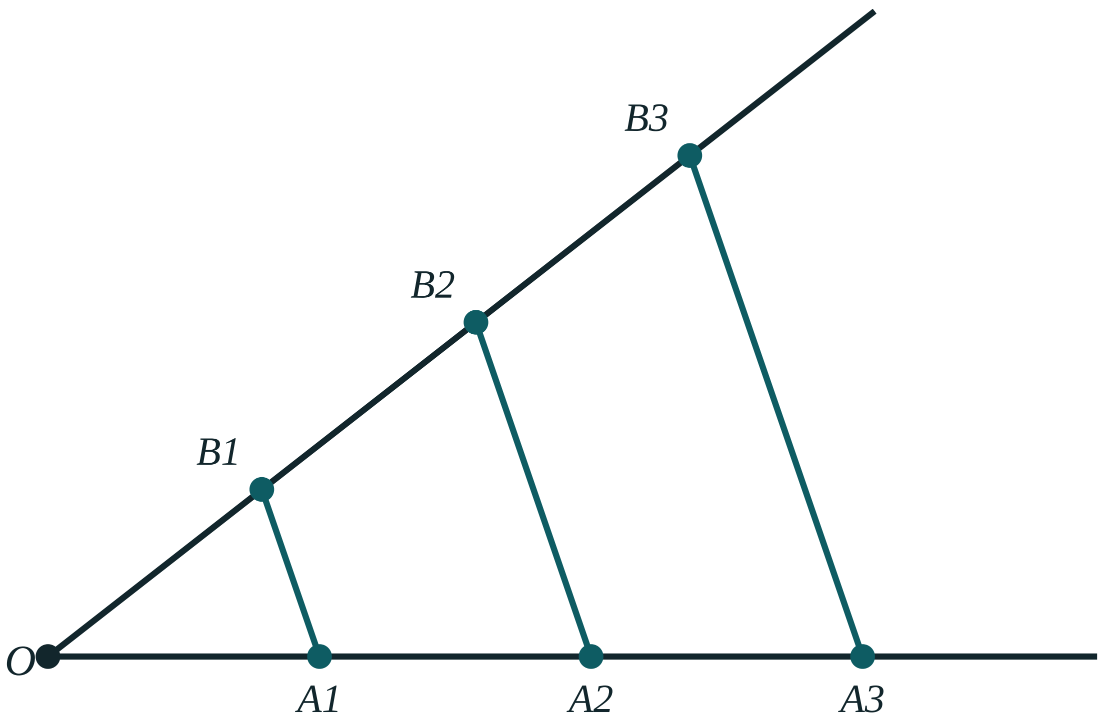
ANSWER

$15\ m$

03 Interactive model

Measure the classroom window with a metre rule and a shadow; the criterion becomes an instrument rather than a theorem.

Thales' theorem



OA_1 **88**	A_1A_2 **88**	OB_1 **88**	B_1B_2 **88**	$OA_1 : OA_2$ **1 : 2**	$OB_1 : OB_2$ **1 : 2**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Criterion	Alomat	Признак
Two angles	Ikki burchak	Два угла
Corresponding angles	Mos burchaklar	Соответственные углы
Vertical angles	Vertikal burchaklar	Вертикальные углы
Common angle	Umumiy burchak	Общий угол
Parallel	Parallel	Параллельный
Transversal	Kesuvchi	Секущая
Proof	Isbot	Доказательство

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The AA criterion needs:

- one angle
- two angles
- three angles
- two sides

The proof rests on:

- Pythagoras
- the angle sum
- the area formula
- the sine rule

$DE \parallel BC$ gives equal angles by:

- vertical angles
- corresponding angles
- Pythagoras
- chance

In the Λ figure the ratio uses:

- DB
- AB
- DE
- AC only

Angles $50°$, $60°$ and $60°$, $70°$ give:

- similar
- not similar
- congruent
- not enough information

Shadow problems use:

- SSS
- SAS
- AA
- Pythagoras

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Angles *40°*, *60°* and *40°*, *60°*

   ANSWER Similar

2. Angles *30°*, *90°* and *60°*, *90°*

   ANSWER Similar

3. Angles *50°*, *60°* and *50°*, *80°*

   ANSWER Not similar

4. $DE \parallel BC$, $AD = 3$, $DB = 3$: k

   ANSWER 2

5. $AD = 4$, $DB = 6$: AB

   ANSWER 10

6. Same: k

   ANSWER 2.5

7. $DE = 5$, $k = 2.5$: BC

   ANSWER 12.5

Medium · the standard question

7 PROBLEMS

1. $AD = 4, DB = 6, DE = 5$: BC

ANSWER 12.5

2. $AD = 5, AB = 15, BC = 21$: DE

ANSWER 7

3. A $1.8\ m$ person, $2.4\ m$ shadow; tower shadow $20\ m$

ANSWER $15\ m$

4. A $2\ m$ pole, $3\ m$ shadow; tree shadow $18\ m$

ANSWER $12\ m$

5. $DE \parallel BC, AD : DB = 2 : 3$: $DE : BC$

ANSWER $2 : 5$

6. Same: ratio of the areas of $\triangle ADE$ and $\triangle ABC$

ANSWER $4 : 25$

7. Two right triangles share an acute angle: are they similar?

ANSWER Yes

Hard · stretch and reason

7 PROBLEMS

1. $DE \parallel BC$, $\triangle ADE$ has area 9 and $\triangle ABC$ has 25: $AD : AB$

   ANSWER $3 : 5$

2. Same figure: $AD : DB$

   ANSWER $3 : 2$

3. The area of the trapezium $DBCE$

   ANSWER 16

4. In the X figure with $AB \parallel CD$, $AB = 6$, $CD = 9$, $AO = 4$: OD

   ANSWER 6

5. A mirror on the ground $3\ m$ from a $1.5\ m$ observer and $12\ m$ from a tower

   ANSWER $6\ m$

6. A triangle is divided by a line parallel to the base into equal areas: the ratio of the parts of a side

   ANSWER $1 : (\sqrt{2} - 1)$

7. Two triangles have angles x, $2x$ and $2x$, $3x$, $x = 30°$: similar?

   ANSWER Yes — both are $\{30°, 60°, 90°\}$

07 Homework

Homework — 5 tasks

State the two equal angles by name before claiming similarity.

1. In $\triangle ABC$, $DE \parallel BC$, $AD = 6$, $DB = 4$, $DE = 9$. Find BC.
2. A $1.6\ m$ person casts a $2\ m$ shadow; a building casts a $35\ m$ shadow. Find its height.
3. Two triangles have angles $45°$, $55°$ and $55°$, $80°$. Are they similar?
4. In the X figure with $AB \parallel CD$, $AB = 8$, $CD = 12$, $AO = 6$. Find OD.

5. Prove the AA criterion in two lines.

The second criterion of similarity — two sides and the included angle

When only one angle is known, two proportional sides around it finish the job.

LESSON 6 | 1 × 40 MIN | GEOMETRIYA 9, §4 | IGX 11.2

LESSON CLOCK

10' | 12' | 12' | 6'

The criterion	10 min
The standard figure	12 min
Why “included” matters	12 min
Homework	6 min

LEARNING OBJECTIVES

- State the SAS similarity criterion precisely, including the word “included”.
- Apply it to a figure with a common angle and two ratios.
- Distinguish it from the congruence criterion of the same name.
- Show by a counter-example that the angle must be the included one.

TEXTBOOK

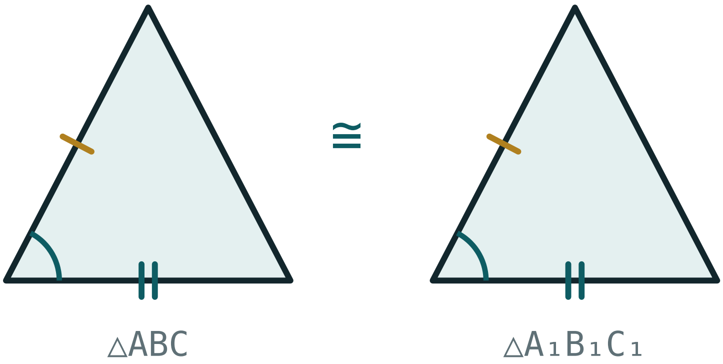
National	pp. 24–27
Cambridge	Core & Extended, pp. 226–231
Workbook	Exercise 11.2

01 Explanation

The criterion

> If two sides of one triangle are proportional to two sides of another, and the angles between those sides are equal, the triangles are similar

$$\frac{\mathrm{AB}}{\mathrm{A_1B_1}} = \frac{\mathrm{AC}}{\mathrm{A_1C_1}} \text{ and } \angle A = \angle A_1 \Longrightarrow \triangle ABC \sim \triangle A_1B_1C_1$$



Two sides and the angle between them fix the shape — proportionally, not just exactly.

THE SAME CRITERION AS FOR CONGRUENCE, ONE WORD CHANGED

Congruence: two sides **equal** and the included angle equal. Similarity: two sides **proportional** and the included angle equal. Every congruence criterion has a similarity twin of exactly this form.

The standard figure

The criterion is used almost exclusively when the two triangles **share** an angle.

Example. In $\triangle ABC$, take D on AB and E on AC with $AD = 3$, $AB = 12$, $AE = 4$, $AC = 16$. Then

$$\frac{AD}{AB} = \frac{3}{12} = \frac{1}{4} \text{ and } \frac{AE}{AC} = \frac{4}{16} = \frac{1}{4}$$

and $\angle A$ is common, so $\triangle ADE \sim \triangle ABC$ with $k = 4$ — **without** being told that $DE \parallel BC$. Indeed the similarity now proves the parallelism.

THE CRITERION RUNS THE Λ FIGURE BACKWARDS

In the last lesson, parallel lines gave the ratios. Here the ratios give the parallel lines. Which direction a question is going is worth deciding before starting.

Why “included” matters

The equal angle must lie **between** the two proportional sides. Otherwise the claim is false.

GIVEN	SIMILAR?
$\dfrac{AB}{A_1B_1} = \dfrac{AC}{A_1C_1}$, $\angle A = \angle A_1$	yes — included
$\dfrac{AB}{A_1B_1} = \dfrac{AC}{A_1C_1}$, $\angle B = \angle B_1$	not necessarily

The second row is the “ambiguous case” familiar from congruence: two sides and a non-included angle can produce two genuinely different triangles.

> CHECK WHERE THE ANGLE SITS BEFORE INVOKING THE CRITERION
>
> A proof that says “two sides proportional and an angle equal” without saying **which** angle is incomplete, and in an examination it is marked as incomplete.

02 Worked examples

EXAMPLE 1 **In $\triangle ABC$, $AD = 3$, $AB = 12$, $AE = 4$, $AC = 16$. Show that $\triangle ADE \sim \triangle ABC$ and find k.**

1. $$\frac{AD}{AB} = \frac{1}{4}$$

2. $$\frac{AE}{AC} = \frac{1}{4}$$

   Equal ratios.

3. $\angle A$ is common — the included angle.

   SAS.

4. $\triangle ADE \sim \triangle ABC$, $k = 4$.

> ANSWER
>
> Similar, $k = 4$

EXAMPLE 2 **In the same figure $DE = 5$. Find BC.**

1. $k = 4$ from the previous part.
2. $BC = DE \times k$
3. $= 5 \times 4$
4. $= 20$

ANSWER

20

EXAMPLE 3 **Two triangles have sides $4, 6$ and $6, 9$ with the included angle $50°$ in each. Are they similar?**

1. $\frac{4}{6} = \frac{2}{3}$
2. $\frac{6}{9} = \frac{2}{3}$

   Equal ratios.
3. The equal angle is included in both.
4. Yes — similar with $k = 1.5$.

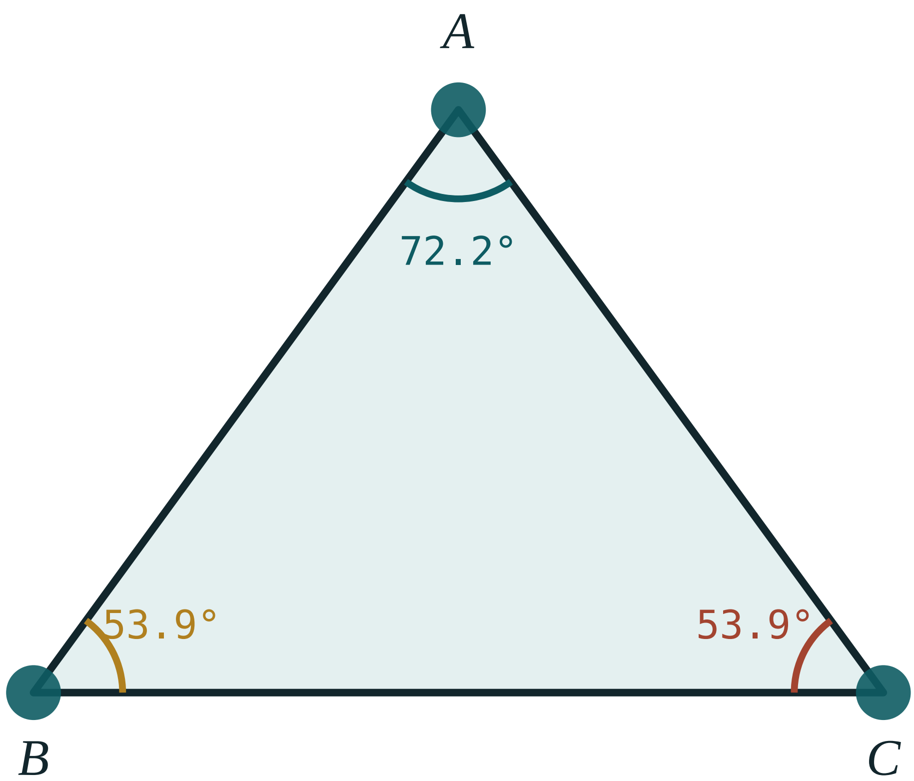
ANSWER

Yes, $k = 1.5$

03 Interactive model

Draw two triangles with the same two ratios but the equal angle in the wrong place, and let the class see that they are not similar; the word “included” then means something.

Two sides and the angle between



∠A **72.2°**	∠B **53.9°**	∠C **53.9°**	sum **180°**	longest side **BC**	largest angle **∠A**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Included angle	Orasidagi burchak	Угол между сторонами
Two sides	Ikki tomon	Две стороны
Proportional	Proporsional	Пропорциональные
Common angle	Umumiy burchak	Общий угол
Second criterion	Ikkinchi alomat	Второй признак
Counter-example	Qarama-qarshi misol	Контрпример
Configuration	Chizma	Конфигурация
Follows	Kelib chiqadi	Следует

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The second criterion needs:

- three sides
- two angles
- two sides and the included angle
- one side

“Included” means the angle is:

- the largest
- between the two sides
- opposite the longest
- anywhere

$AD : AB = AE : AC$ with $\angle A$ common gives:

- nothing
- $\triangle ADE \sim \triangle ABC$
- congruence
- equal areas

It also proves:

- $DE \perp BC$
- $DE \parallel BC$
- $DE = BC$
- nothing

Sides $4, 6$ and $6, 9$, included angle equal:

- similar
- not similar
- congruent
- not enough

A non-included equal angle gives:

- similarity
- congruence
- not necessarily similarity
- a contradiction

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $AD = 3, AB = 12$: ratio

   ANSWER $\frac{1}{4}$

2. $AE = 4, AC = 16$: ratio

   ANSWER $\frac{1}{4}$

3. Common angle plus equal ratios gives

   ANSWER Similarity

4. k in that figure

   ANSWER 4

5. $DE = 5, k = 4$: BC

   ANSWER 20

6. Sides $4, 6$ and $6, 9$, included angle equal

   ANSWER Similar

7. Sides $4, 6$ and $8, 10$, included angle equal

   ANSWER Not similar

Medium · the standard question

7 PROBLEMS

1. $AD = 2, AB = 6, AE = 3, AC = 9$: k

   ANSWER 3

2. Same, $DE = 4$: BC

   ANSWER 12

3. $AD = 5, AB = 20, DE = 6$ with matching ratio on AC: BC

   ANSWER 24

4. Sides $3, 5$ and $9, 15$, included angle equal: k

   ANSWER 3

5. Does the second criterion also prove $DE \parallel BC$?

   ANSWER Yes

6. Two triangles: $6, 8$ and $9, 12$, included $40°$

   ANSWER Similar, $k = 1.5$

7. Two triangles: $6, 8$ and $9, 12$, $40°$ not included

   ANSWER Not necessarily

Hard · stretch and reason

7 PROBLEMS

1. $AD = 4, DB = 8, AE = 5$: EC if $DE \parallel BC$

   ANSWER 10

2. $\triangle ADE \sim \triangle ABC$ with $k = 3$ and $[\triangle ADE] = 7$: $[\triangle ABC]$

   ANSWER 63

3. A triangle with sides $6, 9, 12$ and one with $4, 6, 8$: k

   ANSWER 1.5

4. Are they similar by the second criterion or the third?

   ANSWER The third — all three sides

5. In $\triangle ABC$, $AD = 2, AB = 8, AE = 3, AC = 12$: $[\triangle ADE] : [\triangle ABC]$

   ANSWER $1 : 16$

6. A triangle $5, 7$ with included $60°$ and one $10, 14$ with included $60°$

   ANSWER Similar, $k = 2$

7. Why does the ambiguous case not arise when the angle is included?

   ANSWER Two sides and the angle between them fix the triangle

07 Homework

Homework — 5 tasks

Name the included angle explicitly in every proof.

1. In $\triangle ABC$, $AD = 4$, $AB = 16$, $AE = 5$, $AC = 20$. Prove $\triangle ADE \sim \triangle ABC$ and find k.
2. In the same figure $DE = 6$. Find BC.
3. Two triangles have sides $5, 8$ and $10, 16$ with equal included angles. Find k.
4. Explain why the equal angle must be the included one.
5. Show that the second criterion also proves $DE \parallel BC$ in the standard figure.

The third criterion of similarity — three sides

No angle needed at all: three proportional sides make the same shape.

LESSON 7 | 1 × 40 MIN | GEOMETRIYA 9, §5 | IGX 11.2

LESSON CLOCK

10' | 12' | 12' | 6'

The criterion	10 min
Ordering first	12 min
Choosing a criterion	12 min
Homework	6 min

LEARNING OBJECTIVES

- State the SSS similarity criterion.
- Test three ratios and decide similarity from numbers alone.
- Order the sides before comparing, so that corresponding sides are paired.
- Choose between the three criteria according to what is given.

TEXTBOOK

National	pp. 28–31
Cambridge	Core & Extended, pp. 226–231
Workbook	Exercise 11.2

01 Explanation

The criterion

If the three sides of one triangle are proportional to the three sides of another, the triangles are similar

$$\frac{\mathrm{AB}}{\mathrm{A_1B_1}} = \frac{\mathrm{BC}}{\mathrm{B_1C_1}} = \frac{\mathrm{CA}}{\mathrm{C_1A_1}} \Longrightarrow \triangle ABC \sim \triangle A_1B_1C_1$$

This is the criterion for the case where no angle is known — a set of measurements, and nothing else. It is also the one that a computer would use, because it needs no reasoning about the figure.

THREE NUMBERS, THREE NUMBERS, ONE DECISION

Divide the sides in matching order and look at the three quotients. If they agree, the triangles are similar and the quotient is k; if any one differs, they are not.

Ordering first

Corresponding sides must be paired correctly, and without a figure the only reliable way is to sort both lists.

Example. $\triangle ABC$ has sides $4, 7, 5$ and $\triangle DEF$ has $10.5, 6, 7.5$. Sorted: $4, 5, 7$ and $6, 7.5, 10.5$.

PAIR	RATIO
$6 : 4$	1.5
$7.5 : 5$	1.5
$10.5 : 7$	1.5

All three agree, so the triangles are similar with $k = 1.5$.

COMPARING THE LISTS IN THE ORDER GIVEN IS A LOTTERY

Pairing 4 with 10.5 produces 2.625, and the question looks as if the answer is “not similar”. Sorting both lists takes five seconds and removes the whole risk.

Choosing a criterion

WHAT YOU ARE GIVEN	USE
parallel lines, or a shared angle plus another equal angle	the first (AA)
a shared angle and two ratios around it	the second (SAS)
six side lengths and no angles	the third (SSS)

In practice the first criterion does most of the work, because angles are what diagrams supply. The third is for numerical questions, and the second for the Λ figure run backwards.

SAY WHICH CRITERION YOU USED

A proof that establishes similarity without naming the criterion is incomplete. One sentence — “by the third criterion, since all three ratios equal 1.5” — is what the mark is for.

02 Worked examples

EXAMPLE 1 **Are triangles with sides $4, 7, 5$ and $10.5, 6, 7.5$ similar?**

1. Sort: $4, 5, 7$ and $6, 7.5, 10.5$.
2. $\frac{6}{4} = 1.5$
3. $\frac{7.5}{5} = 1.5$, $\frac{10.5}{7} = 1.5$.
4. Yes — by the third criterion, $k = 1.5$.

ANSWER

Yes, $k = 1.5$

EXAMPLE 2 **Are triangles with sides $3, 4, 5$ and $6, 8, 11$ similar?**

1. $\frac{6}{3} = 2$
2. $\frac{8}{4} = 2$
3. $\frac{11}{5} = 2.2$ — different.
4. No.

   One failing ratio is enough.

ANSWER

No

EXAMPLE 3 **A triangle has sides $6, 8, 10$. A similar triangle has perimeter 36. Find its sides.**

1. Perimeter of the first: 24.
2. $k = \dfrac{36}{24} = 1.5$
3. $6 \times 1.5 = 9$, $8 \times 1.5 = 12$, $10 \times 1.5 = 15$.
4. Check: $9 + 12 + 15 = 36$ ✓

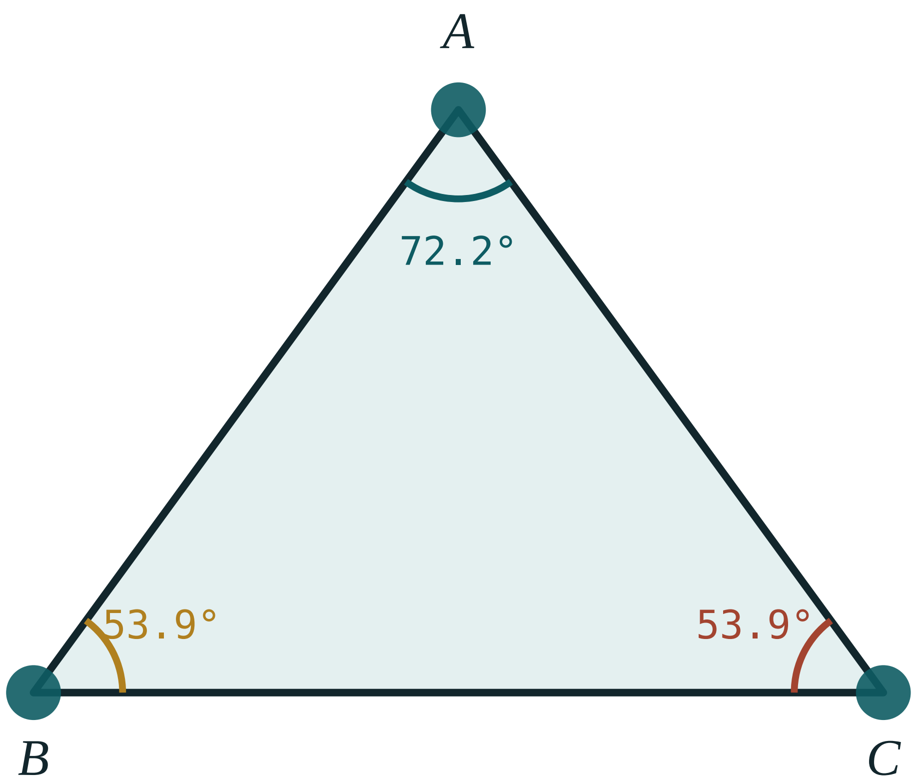
ANSWER

$9, 12, 15$

03 Interactive model

Give the six numbers in scrambled order and let half the class compare as given while the other half sorts first; the disagreement teaches the rule.

Three sides fix the shape



∠A **72.2°** | ∠B **53.9°** | ∠C **53.9°** | sum **180°** | longest side **BC** | largest angle **∠A**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Third criterion	Uchinchi alomat	Третий признак
Three sides	Uchta tomon	Три стороны
To order	Tartiblash	Упорядочить
Ratio test	Nisbat sinovi	Проверка отношений
Shortest side	Eng kichik tomon	Наименьшая сторона
Longest side	Eng katta tomon	Наибольшая сторона
Equal ratios	Teng nisbatlar	Равные отношения
Decide	Aniqlash	Определить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The third criterion needs:

- two angles
- three sides
- two sides and an angle
- one side

Before comparing, you should:

- draw
- sort both lists
- measure the angles
- nothing

$4, 5, 7$ and $6, 7.5, 10.5$ give k =

- 1.25
- 1.5
- 1.75
- 2

$3, 4, 5$ and $6, 8, 11$ are:

- similar
- not similar
- congruent
- right-angled and so similar

With six sides and no angles, use:

- the first
- the second
- the third
- any

A complete proof must:

- give the answer
- name the criterion
- include a graph
- measure

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $3, 4, 5$ and $6, 8, 10$

ANSWER Similar, $k = 2$

2. $2, 3, 4$ and $4, 6, 8$

ANSWER Similar, $k = 2$

3. $3, 4, 5$ and $6, 8, 11$

ANSWER Not similar

4. $5, 12, 13$ and $10, 24, 26$

ANSWER Similar, $k = 2$

5. Ratio $7.5 : 5$

ANSWER 1.5

6. Ratio $10.5 : 7$

ANSWER 1.5

7. Perimeter 24 scaled to 36: k

ANSWER 1.5

Medium · the standard question

7 PROBLEMS

1. $4, 7, 5$ and $10.5, 6, 7.5$

   ANSWER Similar, $k = 1.5$

2. Sides $6, 8, 10$, similar perimeter 36: the sides

   ANSWER $9, 12, 15$

3. $6, 9, 12$ and $8, 12, 16$

   ANSWER Similar, $k = \frac{4}{3}$

4. $4, 6, 8$ and $6, 9, 11$

   ANSWER Not similar

5. Sides $5, 7, 9$ and $15, x, y$

   ANSWER $x = 21, y = 27$

6. Which criterion for six side lengths?

   ANSWER The third

7. Which criterion for two parallel lines?

   ANSWER The first

Hard · stretch and reason

7 PROBLEMS

1. Sides a, $a + 2$, $a + 4$ and $2a$, $2a + 4$, $2a + 8$: similar?

   ANSWER Yes, $k = 2$

2. Two similar triangles have perimeters 20 and 35 and the smaller has sides 4, 7, 9

   ANSWER 7, 12.25, 15.75

3. A triangle 9, 12, 15 is similar to one of area 24: the smaller sides

   ANSWER 6, 8, 10

4. For which x are 4, 6, 8 and 6, 9, x similar?

   ANSWER $x = 12$

5. Are 2, 3, 4 and 3, 4, 5 similar?

   ANSWER No

6. A triangle with sides in the ratio $3 : 4 : 5$ and perimeter 60

   ANSWER 15, 20, 25

7. Its area

   ANSWER 150

07 Homework

Homework — 5 tasks

Sort both lists of sides before comparing, every time.

1. Are triangles with sides 6, 9, 12 and 8, 12, 16 similar?
2. Are triangles with sides 5, 6, 7 and 10, 12, 15 similar?
3. A triangle has sides 8, 15, 17. A similar triangle has perimeter 80. Find its sides.
4. For which x are 3, 5, 7 and 9, 15, x similar?
5. Say which criterion you would use for each of the three standard situations.

Criteria of similarity of right-angled triangles

With one right angle given free, a single further fact is enough.

LESSON 8 | 1 × 40 MIN | GEOMETRIYA 9, §6 | IGX 11.1–11.2

LESSON CLOCK

10' | 14' | 10' | 6'

The three special criteria	10 min
The altitude figure	14 min
Working with it	10 min
Homework	6 min

LEARNING OBJECTIVES

- State the special criteria for right-angled triangles.
- Use one acute angle, or two proportional legs, to establish similarity.
- Use the hypotenuse-and-leg criterion.
- Recognise the three similar triangles created by the altitude to the hypotenuse.

TEXTBOOK

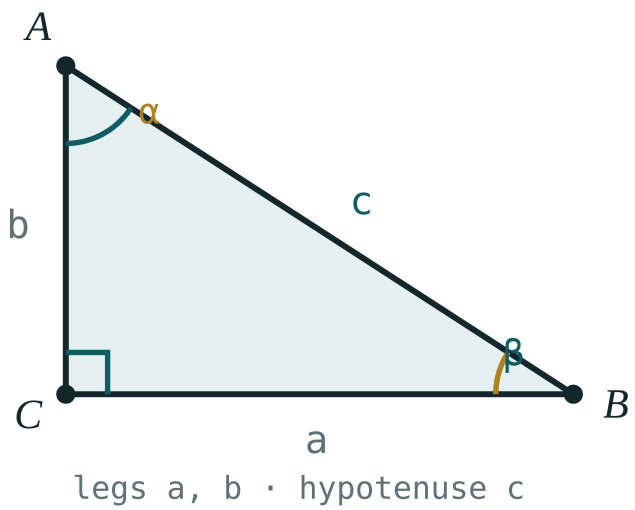
National	pp. 32–35
Cambridge	Core & Extended, pp. 220–231
Workbook	Exercise 11.1

01 Explanation

The three special criteria

One right angle is common to both triangles already, so each general criterion needs one fact fewer.

FROM	NEEDS ONLY	BECAUSE
the first	one equal acute angle	the right angles are the second pair
the second	two proportional legs	the right angle is included
the third	a leg and the hypotenuse proportional	Pythagoras supplies the third side



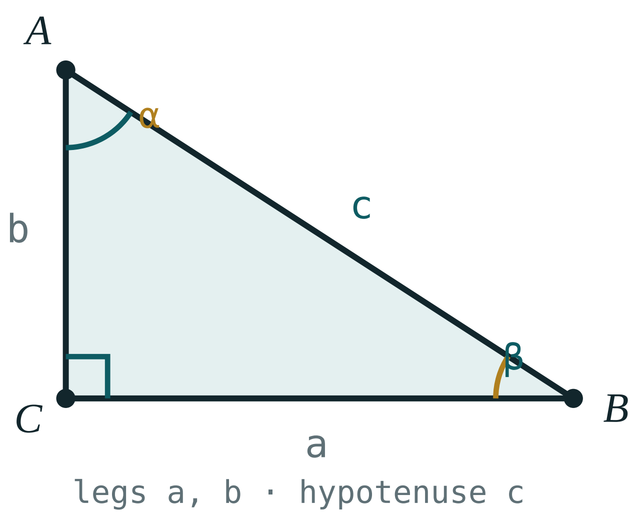
One right angle is free; a single further fact decides similarity.

TWO RIGHT TRIANGLES SHARING AN ACUTE ANGLE ARE ALWAYS SIMILAR

This is the workhorse of the whole chapter, and the reason shadow problems, ladder problems and the altitude figure below all work.

The altitude figure

Drop the altitude CH from the right angle C to the hypotenuse AB. Three triangles appear — $\triangle ACH$, $\triangle CBH$ and $\triangle ABC$ — and all three are similar.



The altitude from the right angle splits the triangle into two copies of itself.

PAIR	EQUAL ANGLES
$\triangle ACH \sim \triangle ABC$	$\angle A$ *common, both right-angled*
$\triangle CBH \sim \triangle ABC$	$\angle B$ *common, both right-angled*
$\triangle ACH \sim \triangle CBH$	both similar to $\triangle ABC$

Reading the proportions off gives three relations, which Quarter IV will use constantly:

$$CH^2 = AH \cdot HB \quad AC^2 = AH \cdot AB \quad BC^2 = BH \cdot AB$$

EACH LEG IS THE MEAN PROPORTIONAL BETWEEN THE HYPOTENUSE AND ITS OWN PROJECTION

And the altitude is the mean proportional between the two projections. Three sentences, three formulae — and adding the last two gives Pythagoras' theorem in one line.

Working with it

Example. In a right triangle the altitude to the hypotenuse divides it into 4 and 9. Find the altitude and the two legs.

QUANTITY	RELATION	VALUE
CH	$CH^2 = 4 \times 9$	6
AC	$AC^2 = 4 \times 13$	$2\sqrt{13}$
BC	$BC^2 = 9 \times 13$	$3\sqrt{13}$

Check: $52 + 117 = 169 = 13^2$ ✓ — Pythagoras holds, as it must.

AB IS 13, NOT 9

The relation for a leg uses the **whole** hypotenuse and that leg's own projection. Using the other projection, or only part of the hypotenuse, is the usual slip here.

02 Worked examples

EXAMPLE 1 **The altitude to the hypotenuse divides it into *4* and *9*. Find the altitude.**

1. $\triangle ACH \sim \triangle CBH$

   Both similar to the whole.

2. $\dfrac{\mathrm{AH}}{\mathrm{CH}} = \dfrac{\mathrm{CH}}{\mathrm{HB}}$

3. $CH^2 = 4 \times 9 = 36$

4. $CH = 6$

ANSWER

6

EXAMPLE 2 **In the same triangle find the two legs.**

1. $AB = 4 + 9 = 13$

   The whole hypotenuse.

2. $AC^2 = AH \cdot AB = 4 \times 13 = 52$

3. $AC = 2\sqrt{13}$

4. $BC^2 = 9 \times 13 = 117$, so $BC = 3\sqrt{13}$.

ANSWER

$2\sqrt{13}$ and $3\sqrt{13}$

EXAMPLE 3 **Two right triangles have legs $6, 8$ and $9, 12$. Are they similar?**

1. $\frac{9}{6} = 1.5$

2. $\frac{12}{8} = 1.5$

   Equal ratios.

3. The right angle is the included angle.

4. Yes — by the second criterion, $k = 1.5$.

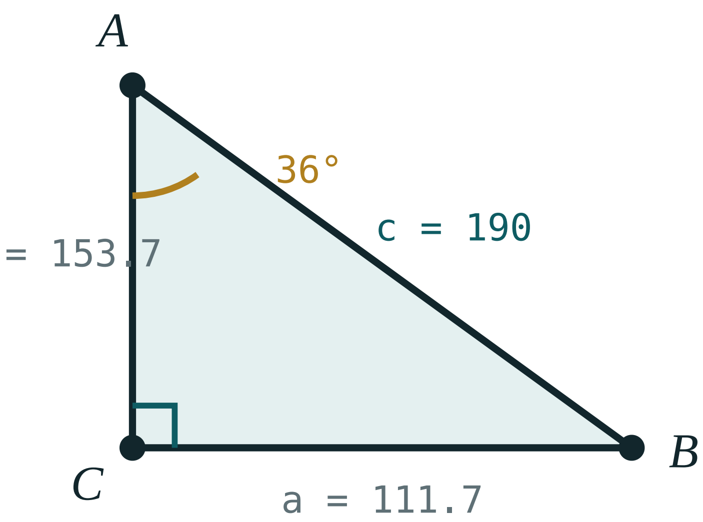
ANSWER

Yes, $k = 1.5$

03 Interactive model

Cut a paper right triangle along the altitude from the right angle; the two pieces can be laid on the original to show all three are the same shape.

The altitude to the hypotenuse



α **36°** | β **54°** | a (opposite) **111.7** | b (adjacent) **153.7** | c (hyp) **190** | sin α **0.59**

cos α **0.81** | tan α **0.73** | $a^2 + b^2$ **36100** | c^2 **36100**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Right-angled triangle	To‘g‘ri burchakli uchburchak	Прямоугольный треугольник
Leg	Katet	Катет
Hypotenuse	Gipotenuza	Гипотенуза
Acute angle	O‘tkir burchak	Острый угол
Altitude	Balandlik	Высота
Foot of the altitude	Balandlik asosi	Основание высоты
Projection	Proyeksiya	Проекция
Mean proportional	O‘rta proporsional	Среднее пропорциональное

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Two right triangles with one equal acute angle are:

- congruent
- similar
- equal in area
- unrelated

The altitude to the hypotenuse creates:

- two similar triangles
- three similar triangles
- no similar triangles
- congruent triangles

CH^2 equals:

- $AH \cdot AB$
- $AH \cdot HB$
- $HB \cdot AB$
- AB^2

AC^2 equals:

- $AH \cdot HB$
- $AH \cdot AB$
- $HB \cdot AB$
- CH^2

Projections 4 and 9 give an altitude of:

- 5
- 6
- 6.5
- 13

Adding the two leg relations gives:

- the sine rule
- Pythagoras
- the area
- nothing

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Projections 4 and 9: the altitude

   ANSWER 6

2. Projections 9 and 16: the altitude

   ANSWER 12

3. Projections 3 and 12: the altitude

   ANSWER 6

4. CH^2 equals

   ANSWER $AH \cdot HB$

5. AC^2 equals

   ANSWER $AH \cdot AB$

6. Legs $6, 8$ and $9, 12$: similar?

   ANSWER Yes, $k = 1.5$

7. Two right triangles with a $30°$ angle each

   ANSWER Similar

Medium · the standard question

7 PROBLEMS

1. Projections 4 and 9: the legs

ANSWER $2\sqrt{13}$ and $3\sqrt{13}$

2. Projections 9 and 16: the legs

ANSWER 15 and 20

3. Legs 15 and 20: the hypotenuse

ANSWER 25

4. Hypotenuse 25, one projection 9: that leg

ANSWER 15

5. Altitude 12, one projection 9: the other

ANSWER 16

6. A 3–4–5 triangle: the altitude to the hypotenuse

ANSWER 2.4

7. Its two projections

ANSWER 1.8 and 3.2

Hard · stretch and reason

7 PROBLEMS

1. A right triangle with hypotenuse 13 and altitude 6: the projections

   ANSWER 4 and 9

2. Legs a and b: the altitude to the hypotenuse

   ANSWER $\dfrac{ab}{\sqrt{a^2 + b^2}}$

3. A $5–12–13$ triangle: its altitude to the hypotenuse

   ANSWER $\dfrac{60}{13}$

4. Its two projections

   ANSWER $\dfrac{25}{13}$ and $\dfrac{144}{13}$

5. Prove Pythagoras from the two leg relations

   ANSWER $AH \cdot AB + HB \cdot AB = AB^2$

6. A right triangle: the altitude is 6 and one projection twice the other

   ANSWER $AH = 3\sqrt{2}$, $HB = 6\sqrt{2}$

7. A ladder $13\ m$ reaches $12\ m$ up a wall: its foot from the wall

   ANSWER $5\ m$

07 Homework

Homework — 5 tasks

Label the foot of the altitude on every figure; the three relations are unreadable without it.

1. The altitude to the hypotenuse divides it into 9 and 16. Find the altitude and the legs.
2. A right triangle has legs 9 and 12. Find the altitude to the hypotenuse.
3. Find the projections of the legs in that triangle.
4. Two right triangles have legs 5, 12 and 10, 24. Are they similar?

5. Derive Pythagoras’ theorem from the two leg relations.

Applying the criteria to problems on proof

Similarity is mostly used not to find lengths but to prove that two lines are parallel, or two products equal.

LESSON 9 | 1 × 40 MIN | GEOMETRIYA 9, §7 | IGX 11.4

LESSON CLOCK

10' | 14' | 10' | 6'

The shape of a proof	10 min
Proportions and products	14 min
The bisector property	10 min
Homework	6 min

LEARNING OBJECTIVES

- Write a proof that uses similarity as its central step.
- Convert a proportion into an equal-products statement and back.
- Prove the bisector property of a triangle.
- Lay a proof out in the standard statement–reason form.

TEXTBOOK

National	pp. 36–40
Cambridge	Core & Extended, pp. 237–241
Workbook	Exercise 11.4

01 Explanation

The shape of a proof

Almost every similarity proof has the same four lines.

LINE	CONTENT
1	name two equal angles (or the three ratios)
2	conclude the similarity, naming the criterion
3	write the proportion that follows
4	rearrange it into what was asked

WRITE REASONS, NOT ONLY STATEMENTS

“$\angle A = \angle A$ (common)”, “$\angle ADE = \angle ABC$ (corresponding angles, $DE \parallel BC$)”. Each bracket is a mark. A chain of true statements without reasons is not a proof.

Proportions and products

Examination questions almost always ask for a product, not a ratio.

$$\frac{a}{b} = \frac{c}{d} \iff ad = bc$$

Example. In $\triangle ABC$, CH is the altitude from the right angle. Prove $AC^2 = AH \cdot AB$.

STATEMENT	REASON
$\angle A = \angle A$	common
$\angle AHC = \angle ACB = 90°$	given
$\triangle AHC \sim \triangle ACB$	first criterion
$\frac{AH}{AC} = \frac{AC}{AB}$	corresponding sides
$AC^2 = AH \cdot AB$	cross-multiplying

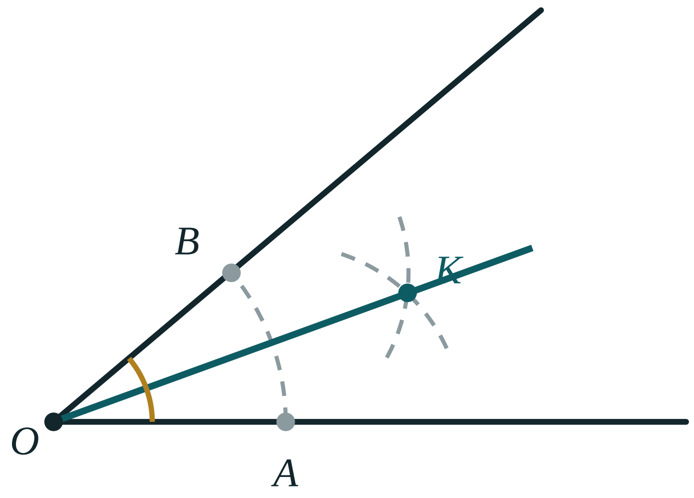
THE MIDDLE TERM REPEATS — THAT IS THE SIGNAL

A statement of the form $x^2 = pq$ always comes from a proportion in which x appears twice. Looking for that repeated term tells you which two triangles to compare.

The bisector property

The bisector of an angle of a triangle divides the opposite side in the ratio of the two adjacent sides

$$\frac{BD}{DC} = \frac{AB}{AC}$$



The bisector from A meets BC at D, cutting it in the ratio of the two sides at A.

The proof draws an auxiliary line: through C, a parallel to AD meeting BA extended at E. Then $\triangle ACE$ is isosceles and the similar triangles $\triangle BAD$ and $\triangle BEC$ give the ratio.

AN AUXILIARY LINE IS A LEGITIMATE MOVE, NOT A TRICK

Many proofs are impossible until one extra line is drawn — usually a parallel, an altitude, or an extension of a side. Trying those three, in that order, solves most problems that appear to have no route in.

02 Worked examples

EXAMPLE 1 **Prove $AC^2 = AH \cdot AB$ where CH is the altitude from the right angle C.**

1. $\angle A$ is common to $\triangle AHC$ and $\triangle ACB$.
2. $\angle AHC = \angle ACB = 90°$.
3. $\triangle AHC \sim \triangle ACB$ by the first criterion.
4. $\dfrac{AH}{AC} = \dfrac{AC}{AB} \Rightarrow AC^2 = AH \cdot AB$.

ANSWER

Proved

EXAMPLE 2 **In $\triangle ABC$, $AB = 8$, $AC = 12$ and the bisector from A meets BC at D. If $BC = 15$, find BD.**

1. $\dfrac{BD}{DC} = \dfrac{8}{12} = \dfrac{2}{3}$
2. Let $BD = 2t$, $DC = 3t$.
3. $5t = 15 \Rightarrow t = 3$
4. $BD = 6$

ANSWER

$BD = 6$

EXAMPLE 3 **In $\triangle ABC$, D on AB and E on AC satisfy $AD \cdot AB = AE \cdot AC$. Prove $\triangle ADE \sim \triangle ACB$.**

1. Rewrite: $\frac{\mathrm{AD}}{\mathrm{AC}} = \frac{\mathrm{AE}}{\mathrm{AB}}$.

   Note the crossing.

2. $\angle A$ is common.

3. Two sides proportional with the included angle equal.

   Second criterion.

4. $\triangle ADE \sim \triangle ACB$.

   Vertices in that order.

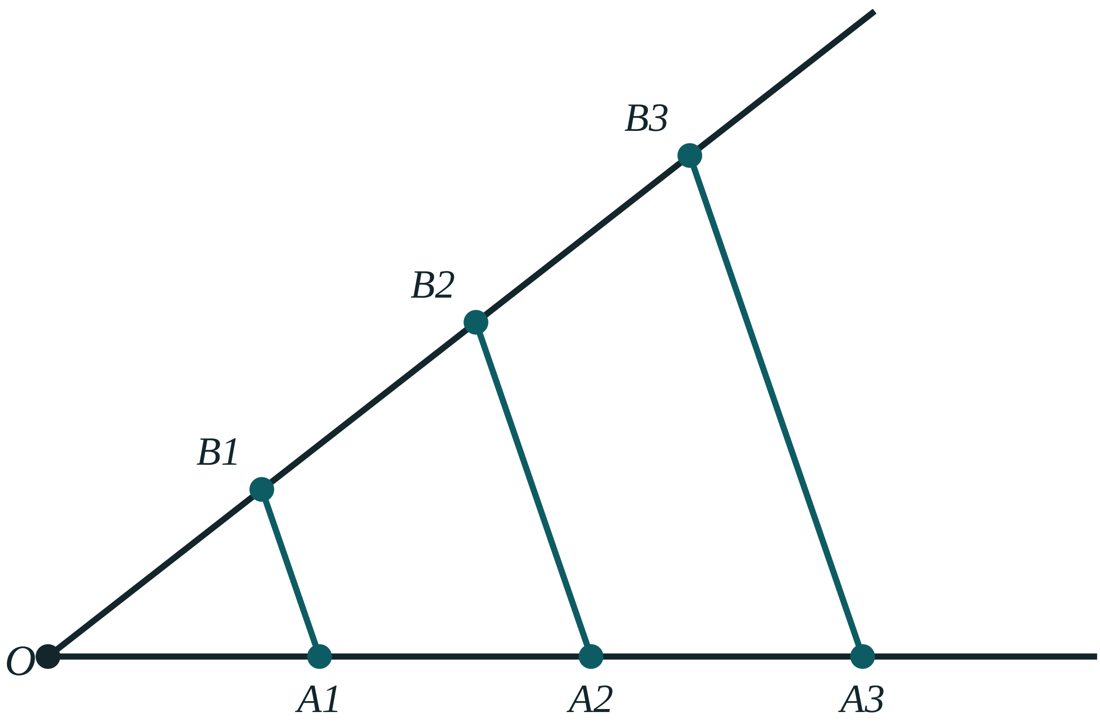
ANSWER

Proved

03 Interactive model

Write a proof on the board with the reasons column blank and ask the class to fill it in; they discover that the reasons are the proof.

Thales' theorem



OA_1 **88**	A_1A_2 **88**	OB_1 **88**	B_1B_2 **88**	$OA_1 : OA_2$ **1 : 2**	$OB_1 : OB_2$ **1 : 2**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Proof	Isbot	Доказательство
Statement	Tasdiq	Утверждение
Reason	Asos	Обоснование
To follow	Kelib chiqmoq	Следовать
Bisector	Bissektrisa	Биссектриса
Equal products	Teng ko‘paytmalar	Равные произведения
Converse	Teskari teorema	Обратная теорема
Auxiliary line	Yordamchi chiziq	Вспомогательная линия

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A similarity proof begins by:

- writing the answer
- naming equal angles
- measuring
- drawing a graph

$\frac{a}{b} = \frac{c}{d}$ is the same as:

- $ac = bd$
- $ad = bc$
- $a + d = b + c$
- $ab = cd$

$x^2 = pq$ comes from a proportion where:

- x appears once
- x appears twice
- $p = q$
- nothing repeats

The bisector from A cuts BC in the ratio:

- $AB : BC$
- $AB : AC$
- $AC : BC$
- $1 : 1$

$AB = 8, AC = 12, BC = 15$: $BD =$

- 5
- 6
- 7.5
- 9

An auxiliary line is:

- cheating
- a legitimate step
- always a circle
- never needed

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Rewrite $\frac{a}{b} = \frac{c}{d}$ as products

ANSWER $ad = bc$

2. Rewrite $x^2 = pq$ as a proportion

ANSWER $\frac{p}{x} = \frac{x}{q}$

3. Bisector ratio in $\triangle ABC$ from A

ANSWER $\frac{AB}{AC}$

4. $AB = 6, AC = 9$: $BD : DC$

ANSWER $2 : 3$

5. $AB = 5, AC = 5$: $BD : DC$

ANSWER $1 : 1$

6. Reason for "$\angle A = \angle A$"

ANSWER Common angle

7. Criterion used with two equal angles

ANSWER The first

Medium · the standard question

7 PROBLEMS

1. $AB = 8$, $AC = 12$, $BC = 15$: BD

   ANSWER 6

2. Same: DC

   ANSWER 9

3. $AB = 10$, $AC = 15$, $BD = 4$: DC

   ANSWER 6

4. Prove $AC^2 = AH \cdot AB$

   ANSWER Similar triangles, then cross-multiply

5. Prove $CH^2 = AH \cdot HB$

   ANSWER $\triangle ACH \sim \triangle CBH$

6. $AD \cdot AB = AE \cdot AC$ with $\angle A$ common gives

   ANSWER $\triangle ADE \sim \triangle ACB$

7. A bisector divides $BC = 21$ in $3 : 4$: the parts

   ANSWER 9 and 12

Hard · stretch and reason

7 PROBLEMS

1. $AB = 6, AC = 8, BC = 7$: BD

ANSWER 3

2. A triangle with a bisector cutting BC into 4 and 6, $AB = 10$: AC

ANSWER 15

3. Prove that $DE \parallel BC$ follows from $\frac{AD}{AB} = \frac{AE}{AC}$

ANSWER Second criterion, then equal corresponding angles

4. Two chords AB and CD meet at P: prove $PA \cdot PB = PC \cdot PD$

ANSWER $\triangle APC \sim \triangle DPB$

5. In that figure $PA = 4, PB = 9, PC = 6$: PD

ANSWER 6

6. A tangent PT and a secant PAB: prove $PT^2 = PA \cdot PB$

ANSWER $\triangle PTA \sim \triangle PBT$

7. $PA = 4, PB = 9$: PT

ANSWER 6

07 Homework

Homework — 5 tasks

Set every proof out in two columns — statement and reason.

1. Prove $BC^2 = BH \cdot AB$ in the altitude figure.
2. In $\triangle ABC$, $AB = 9$, $AC = 12$, $BC = 14$. Find the parts into which the bisector from A divides BC.
3. Prove that a line dividing two sides of a triangle proportionally is parallel to the third.
4. Two chords meet inside a circle at P. Prove $PA \cdot PB = PC \cdot PD$.

5. In that figure $PA = 3$, $PB = 8$, $PC = 4$. Find PD.

Transformations of the plane — movement and translation

A second way to think about figures: not as sets of points but as things that can be moved.

LESSON 10 | 1 × 40 MIN | GEOMETRIYA 9, §8 | IGX 3.1 EXTENSION

LESSON CLOCK

10' | 14' | 10' | 6'

What a transformation is	10 min
Translation	14 min
What is preserved	10 min
Homework	6 min

LEARNING OBJECTIVES

- Define a transformation of the plane and a movement (isometry).
- Describe a translation by a vector and find images of points.
- Know what a movement preserves and what it does not.
- Compose two translations.

TEXTBOOK

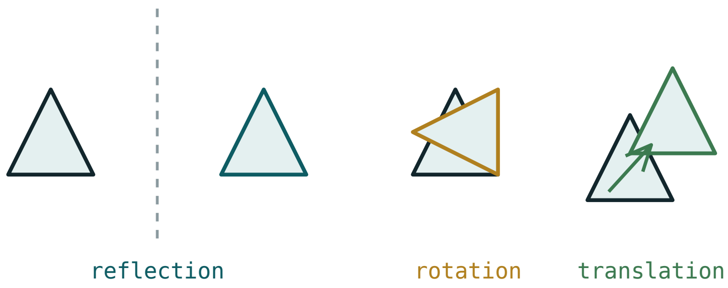
National	pp. 41–45
Cambridge	Core & Extended, pp. 40–46
Workbook	Exercise 3.1

01 Explanation

What a transformation is

A **transformation** of the plane is a rule that sends every point P to exactly one point P', its **image**. A transformation that preserves distance is a **movement** — in Cambridge, an **isometry**.

TRANSFORMATION	A MOVEMENT?	EFFECT
translation	yes	slides the figure
axial symmetry (reflection)	yes	flips it
rotation	yes	turns it
homothety (enlargement)	no	resizes it



The four transformations of the Grade 9 course — three preserve size, one does not.

MOVEMENTS PRODUCE CONGRUENT FIGURES, HOMOTHETY SIMILAR ONES

That single sentence organises the whole chapter: the first three lessons are about congruence in motion, the last three about similarity in motion.

Translation

A **translation** by the vector $\vec{a}(m, n)$ sends $P(x, y)$ to

$$P'(x + m, y + n)$$

Every point moves the same distance in the same direction, so PP' is the same segment for every P — which is exactly what a vector is.

POINT	VECTOR	IMAGE
$A(1, 2)$	$(3, -1)$	$A'(4, 1)$
$B(-2, 5)$	$(3, -1)$	$B'(1, 4)$
$C(0, 0)$	$(3, -1)$	$C'(3, -1)$

Composing two translations gives a third: translate by $(3, -1)$ then by $(1, 4)$ and the result is a translation by $(4, 3)$ — the vectors add.

A TRANSLATION HAS NO FIXED POINTS

Unless the vector is zero, nothing stays where it was. A rotation fixes its centre and a reflection fixes its axis; a translation fixes nothing — a fact that distinguishes it at once.

What is preserved

PRESERVED BY A MOVEMENT	NOT PRESERVED
distances	position
angles	orientation (by a reflection)
areas	—
parallelism	—
straightness	—

Because distances are preserved, a figure and its image under a movement are **congruent**. Under a homothety they are only similar — angles survive but distances are multiplied by k.

CAMBRIDGE ASKS FOR A FULL DESCRIPTION

“A translation” is not an answer; “a translation by the vector $(3, -1)$” is. Each type of transformation has its own required data: a vector, an axis, a centre and angle, or a centre and a scale factor.

02 Worked examples

EXAMPLE 1 **Translate $\triangle ABC$ with $A(1, 2)$, $B(4, 2)$, $C(1, 6)$ by $(3, -1)$.**

1. $A'(1 + 3, 2 - 1) = A'(4, 1)$
2. $B'(7, 1)$
3. $C'(4, 5)$
4. The image is congruent to the original.
   A movement.

ANSWER

$A'(4, 1), B'(7, 1), C'(4, 5)$

EXAMPLE 2 **A translation sends $P(2, 5)$ to $P'(-1, 9)$. Find its vector, and the image of $Q(4, 0)$.**

1. $m = -1 - 2 = -3$
2. $n = 9 - 5 = 4$

   Vector $(-3, 4)$.
3. $Q'(4 - 3, 0 + 4)$
4. $Q'(1, 4)$

ANSWER

Vector $(-3, 4)$; $Q'(1, 4)$

EXAMPLE 3 **Translate by $(3, -1)$ and then by $(1, 4)$. Describe the single transformation that does the same.**

1. Each point gains 3 then 1 in x.
2. And loses 1 then gains 4 in y.
3. Total: $(4, 3)$.

   The vectors add.
4. A translation by $(4, 3)$.

   Full description.

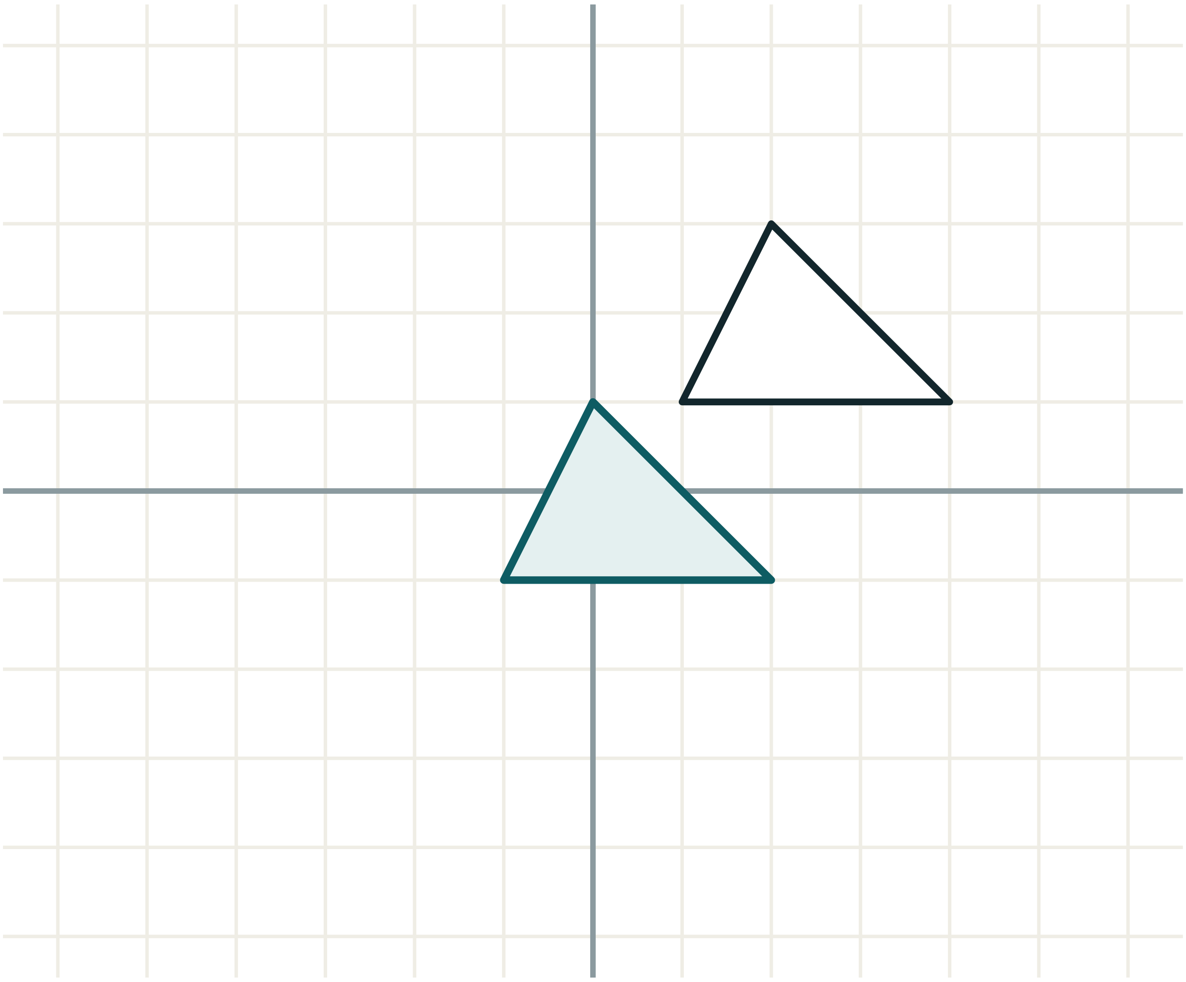
ANSWER

A translation by $(4, 3)$

03 Interactive model

Slide a cut-out triangle across squared paper without turning it; every vertex traces the same arrow, and the vector is that arrow.

Slide the figure



transformation **translation** | lengths **unchanged** | angles **unchanged** | orientation **kept**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Transformation	Almashtirish	Преобразование
Movement	Harakat	Движение
Isometry	Izometriya	Изометрия
Translation	Parallel ko‘chirish	Параллельный перенос
Image	Tasvir	Образ
Pre-image	Asl nusxa	Прообраз
Vector	Vektor	Вектор
Composition	Kompozitsiya	Композиция

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A movement preserves:

- position
- distance
- colour
- nothing

A translation is described by:

- an axis
- a centre
- a vector
- a scale factor

$P(1, 2)$ translated by $(3, -1)$:

- $(4, 1)$
- $(3, 1)$
- $(4, 3)$
- $(-2, 3)$

A translation fixes:

- one point
- a line
- nothing
- everything

Two translations compose into:

- a rotation
- a reflection
- a translation
- nothing

A homothety is:

- a movement
- not a movement
- a reflection
- a translation

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $(1, 2)$ by $(3, -1)$

   ANSWER $(4, 1)$

2. $(0, 0)$ by $(3, -1)$

   ANSWER $(3, -1)$

3. $(-2, 5)$ by $(3, -1)$

   ANSWER $(1, 4)$

4. $(4, 2)$ by $(-1, 3)$

   ANSWER $(3, 5)$

5. Does a translation preserve area?

   ANSWER Yes

6. Does a translation have a fixed point?

   ANSWER No

7. A translation is described by

   ANSWER A vector

Medium · the standard question

7 PROBLEMS

1. $P(2,5) \to P'(-1,9)$: the vector

   ANSWER $(-3, 4)$

2. Same vector applied to $Q(4, 0)$

   ANSWER $(1, 4)$

3. $(3, -1)$ then $(1, 4)$

   ANSWER $(4, 3)$

4. $(5, 2)$ then $(-5, -2)$

   ANSWER The identity

5. Translate $A(1,2)$, $B(4,2)$, $C(1,6)$ by $(3, -1)$

   ANSWER $(4,1)$, $(7,1)$, $(4,5)$

6. The image triangle is

   ANSWER Congruent to the original

7. Which of the four is not a movement?

   ANSWER Homothety

Hard · stretch and reason

7 PROBLEMS

1. A translation maps $(1, 1)$ to $(4, 5)$: the length of the vector

   ANSWER 5

2. The image of the line $y = 2x$ under $(0, 3)$

   ANSWER $y = 2x + 3$

3. The image of $y = x^2$ under $(2, 0)$

   ANSWER $y = (x - 2)^2$

4. A translation maps the circle $x^2 + y^2 = 9$ by $(1, -2)$

   ANSWER $(x-1)^2 + (y+2)^2 = 9$

5. Composing n translations by $(1, 1)$

   ANSWER (n, n)

6. Two translations commute?

   ANSWER Yes — vector addition is commutative

7. A movement that fixes exactly one point is a

   ANSWER Rotation

07 Homework

Homework — 5 tasks

Give a full description of every transformation — the type *and* its data.

1. Translate $A(2, 1)$, $B(5, 1)$, $C(2, 5)$ by $(-2, 3)$.
2. A translation sends $(3, 4)$ to $(0, 8)$. Find its vector and the image of $(1, 1)$.
3. Compose translations by $(2, -3)$ and $(-5, 1)$.
4. Find the image of the line $y = 3x - 1$ under a translation by $(0, 4)$.
5. State what a movement preserves and give one thing it does not.

Axial symmetry

Reflection in a line — the movement that reverses orientation, and the only one that does.

LESSON 11 | 1 × 40 MIN | GEOMETRIYA 9, §9 | EXTENSION BEYOND IGX

LESSON CLOCK

10' | 14' | 10' | 6'

The construction	10 min
Reflection in the coordinate lines	14 min
Axes of symmetry	10 min
Homework	6 min

LEARNING OBJECTIVES

- Construct the image of a point and a figure under reflection in a line.
- Reflect a point in the axes and in the lines y = x and y = −x.
- Find the axes of symmetry of a given figure.
- Know that reflection reverses orientation but preserves distance.

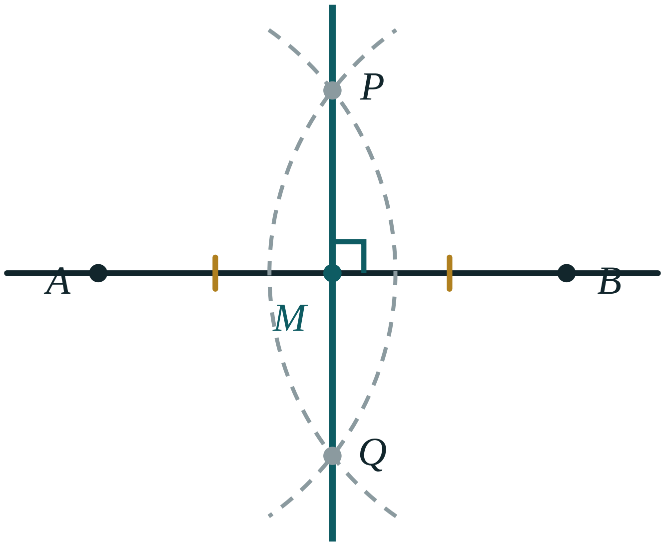
TEXTBOOK

National	pp. 46–50
Cambridge	Core & Extended, pp. 46–50
Workbook	Exercise 3.1

01 Explanation

The construction

To reflect P in the line ℓ: drop a perpendicular from P to ℓ, and continue it the same distance on the other side. The line ℓ is the **perpendicular bisector** of PP'.



The axis is the perpendicular bisector of every segment joining a point to its image.

PROPERTY	STATEMENT
fixed points	every point of ℓ
distance	preserved
angles	preserved in size
orientation	**reversed**
applied twice	the identity

A REFLECTION IS ITS OWN INVERSE

Reflecting twice in the same line returns every point to where it started. No other transformation of the chapter has that property, and it is a useful check on any construction.

Reflection in the coordinate lines

AXIS	$P(X, Y) \mapsto$	EXAMPLE $(3, 5)$
Ox	$(x, -y)$	$(3, -5)$
Oy	$(-x, y)$	$(-3, 5)$
$y = x$	(y, x)	$(5, 3)$
$y = -x$	$(-y, -x)$	$(-5, -3)$
$x = a$	$(2a - x, y)$	$a = 1$: $(-1, 5)$
$y = b$	$(x, 2b - y)$	$b = 2$: $(3, -1)$

The last two rows contain the first four as special cases: $a = 0$ gives reflection in Oy, and $b = 0$ gives reflection in Ox.

> **$Y = X$ SWAPS THE COORDINATES; IT DOES NOT NEGATE THEM**
>
> $(3, 5) \mapsto (5, 3)$, not $(-3, -5)$. Reflecting a graph in $y = x$ is exactly how an inverse function is drawn, which is why this row matters beyond geometry.

Axes of symmetry

A figure has an **axis of symmetry** if reflecting it in that line leaves it unchanged.

FIGURE	AXES
an isosceles triangle	1
an equilateral triangle	3
a rectangle	2
a rhombus	2
a square	4
a regular n-gon	n
a circle	infinitely many
a parallelogram (not a rhombus)	0

A PARALLELOGRAM HAS NO AXIS OF SYMMETRY

Its diagonals are not axes: reflecting in a diagonal does not map the figure to itself. It has **central** symmetry instead — which is the next lesson.

02 Worked examples

EXAMPLE 1 **Reflect $A(3, 5)$ in Ox, in Oy and in $y = x$.**

1. Ox: negate y — $(3, -5)$.
2. Oy: negate x — $(-3, 5)$.
3. $y = x$: swap — $(5, 3)$.
4. Each image is the same distance from the axis.

ANSWER

$(3, -5), (-3, 5), (5, 3)$

EXAMPLE 2 **Reflect $P(3, 5)$ in the line $x = 1$.**

1. Distance from the line: $3 - 1 = 2$.
2. The image is 2 on the other side.
3. $1 - 2 = -1$
4. $P'(-1, 5)$

   y is unchanged.

ANSWER

$(-1, 5)$

EXAMPLE 3 **How many axes of symmetry has a rhombus that is not a square?**

1. Its two diagonals are perpendicular bisectors of each other.
2. Reflecting in either diagonal maps the rhombus to itself.
3. The lines through the midpoints of opposite sides do not.
4. Two.

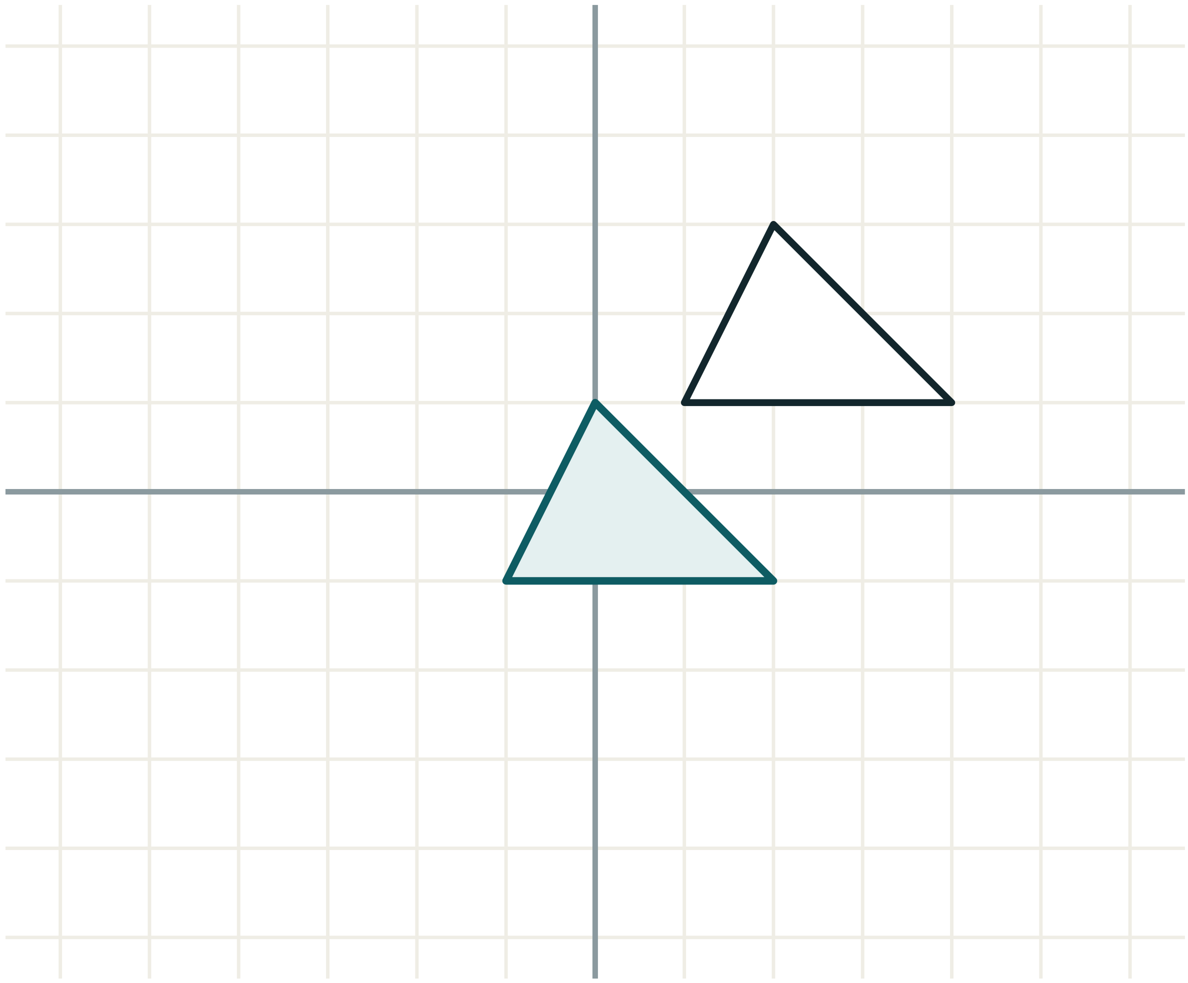
ANSWER

2

03 Interactive model

Fold a paper figure along a proposed axis; if the two halves coincide it is an axis, and if they do not, the class sees exactly why.

Reflect in a line



transformation **translation** | lengths **unchanged** | angles **unchanged** | orientation **kept**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Axial symmetry	O‘qqa nisbatan simmetriya	Осевая симметрия
Reflection	Simmetriya	Отражение
Axis of symmetry	Simmetriya o‘qi	Ось симметрии
Perpendicular bisector	O‘rta perpendikulyar	Серединный перпендикуляр
Orientation	Yo‘nalish	Ориентация
Fixed point	Qo‘zg‘almas nuqta	Неподвижная точка
Mirror line	Ko‘zgu chizig‘i	Зеркальная линия
Symmetric figure	Simmetrik shakl	Симметричная фигура

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

The axis is the _______ of PP':

- midpoint
- perpendicular bisector
- parallel
- extension

Reflection preserves:

- orientation
- distance
- position
- nothing

Reflection reverses:

- distance
- angle size
- orientation
- area

$(3, 5)$ in $y = x$:

- $(-3, -5)$
- $(5, 3)$
- $(3, -5)$
- $(-5, -3)$

A square has how many axes?

- 2
- 3
- 4
- 8

A parallelogram has how many axes?

- 0
- 1
- 2
- 4

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $(3, 5)$ in Ox

   ANSWER $(3, -5)$

2. $(3, 5)$ in Oy

   ANSWER $(-3, 5)$

3. $(3, 5)$ in $y = x$

   ANSWER $(5, 3)$

4. $(-2, 7)$ in Ox

   ANSWER $(-2, -7)$

5. Axes of an equilateral triangle

   ANSWER 3

6. Axes of a rectangle

   ANSWER 2

7. Axes of a circle

   ANSWER Infinitely many

Medium · the standard question

7 PROBLEMS

1. $(3, 5)$ in $x = 1$

   ANSWER $(-1, 5)$

2. $(3, 5)$ in $y = 2$

   ANSWER $(3, -1)$

3. $(3, 5)$ in $y = -x$

   ANSWER $(-5, -3)$

4. Axes of a rhombus (not a square)

   ANSWER 2

5. Axes of a parallelogram

   ANSWER 0

6. Axes of a regular hexagon

   ANSWER 6

7. Reflecting twice in the same line gives

   ANSWER The identity

Hard · stretch and reason

7 PROBLEMS

1. The image of $y = 2x + 1$ in Ox

   ANSWER $y = -2x - 1$

2. The image of $y = 2x + 1$ in Oy

   ANSWER $y = -2x + 1$

3. The image of $y = 2x + 1$ in $y = x$

   ANSWER $y = \dfrac{x - 1}{2}$

4. The image of $y = x^2$ in Ox

   ANSWER $y = -x^2$

5. Reflect (a, b) in $x = c$

   ANSWER $(2c - a, b)$

6. Two reflections in parallel lines d apart give

   ANSWER A translation of $2d$

7. Two reflections in perpendicular lines give

   ANSWER A half-turn about their intersection

07 Homework

Homework — 5 tasks

Draw the axis and the perpendicular for every construction.

1. Reflect $(4, -3)$ in Ox, Oy, $y = x$ and $y = -x$.
2. Reflect $(5, 2)$ in the line $x = 3$.
3. How many axes of symmetry has a regular octagon?
4. Find the image of the line $y = 3x - 2$ under reflection in Oy.

5. Explain why a parallelogram has no axis of symmetry.

Central symmetry and rotation

Turning about a point — and the half-turn, which is the case worth knowing by heart.

LESSON 12 | 1 × 40 MIN | GEOMETRIYA 9, §10 | EXTENSION BEYOND IGX

LESSON CLOCK

10' | 14' | 10' | 6'

Central symmetry	10 min
Rotation	14 min
Rotational symmetry	10 min
Homework	6 min

LEARNING OBJECTIVES

- Define central symmetry as a half-turn about a point.
- Describe a rotation by its centre, angle and direction.
- Find images under a rotation of 90°, 180° or 270° about the origin.
- Identify figures with central symmetry and with rotational symmetry of order n.

TEXTBOOK

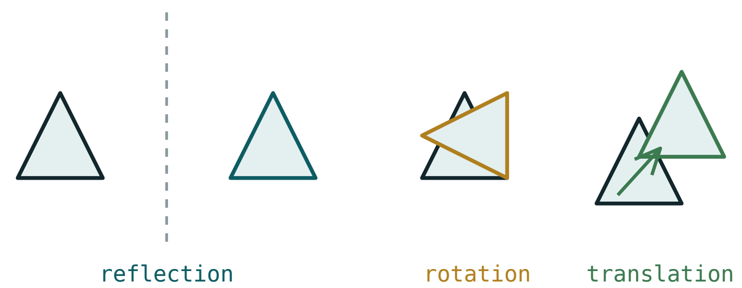
National	pp. 51–56
Cambridge	Core & Extended, pp. 46–50
Workbook	Exercise 3.1

01 Explanation

Central symmetry

The image of P under central symmetry with centre O is the point P' on the line PO with $OP' = OP$ and O between them: O is the **midpoint** of PP'.

> $P(x, y) \mapsto P'(-x, -y)$ for the centre at the origin



A half-turn about O — every point goes through the centre to the far side.

FIGURE	CENTRE OF SYMMETRY?
a parallelogram	yes — the intersection of the diagonals
a circle	yes — its centre
a regular hexagon	yes
an equilateral triangle	**no**
a regular pentagon	**no**

CENTRAL SYMMETRY IS A ROTATION OF $180°$

It is not a separate kind of transformation; it is the special case of a rotation that is worth its own name because it appears so often — in the parallelogram, in the hyperbola, in every odd function.

Rotation

A **rotation** needs three pieces of data: a **centre**, an **angle** and a **direction** (anticlockwise positive, as in trigonometry).

ROTATION ABOUT O	$(X, Y) \mapsto$	EXAMPLE $(3, 1)$
$90°$ anticlockwise	$(-y, x)$	$(-1, 3)$
$180°$	$(-x, -y)$	$(-3, -1)$
$270°$ anticlockwise	$(y, -x)$	$(1, -3)$
$360°$	(x, y)	$(3, 1)$

A rotation of $90°$ clockwise is the same as $270°$ anticlockwise, so the third row answers both.

GIVE ALL THREE PIECES OF DATA

“A rotation of $90°$” is incomplete: about which point, and which way? Cambridge marks the description, and two of the three marks are for the centre and the direction.

Rotational symmetry

A figure has **rotational symmetry of order n** if it maps onto itself n times in a full turn — that is, under a rotation of $\frac{360°}{n}$.

FIGURE	ORDER	SMALLEST ANGLE
an equilateral triangle	3	$120°$
a square	4	$90°$
a rectangle	2	$180°$
a parallelogram	2	$180°$
a regular n-gon	n	$\frac{360°}{n}$
a circle	infinite	any angle

ORDER 2 AND CENTRAL SYMMETRY ARE THE SAME THING

A figure has a centre of symmetry exactly when its rotational symmetry has even order — the half-turn is then one of its symmetries. That is why the parallelogram and the regular hexagon have centres and the equilateral triangle and regular pentagon do not.

02 Worked examples

EXAMPLE 1 **Find the image of $(3, 1)$ under a rotation of $90°$ anticlockwise about the origin.**

1. The rule is $(x, y) \mapsto (-y, x)$.
2. $x = 3, y = 1$
3. $(-1, 3)$
4. Check: both points are $\sqrt{10}$ from the origin ✓

ANSWER

$(-1, 3)$

EXAMPLE 2 **A point $P(4, -2)$ is mapped to $P'(-4, 2)$. Describe the transformation fully.**

1. Both coordinates are negated.
2. That is a half-turn about the origin.
3. Equivalently, central symmetry in O.
4. A rotation of $180°$ about $(0, 0)$.
   Direction is irrelevant at $180°$.

ANSWER

A rotation of $180°$ about the origin

EXAMPLE 3 **Give the order of rotational symmetry and the number of axes for a regular hexagon.**

1. A regular n-gon has order n.
2. Order 6, smallest angle $60°$.
3. It has $n = 6$ axes of symmetry.
4. Order 6 is even, so it also has a centre.

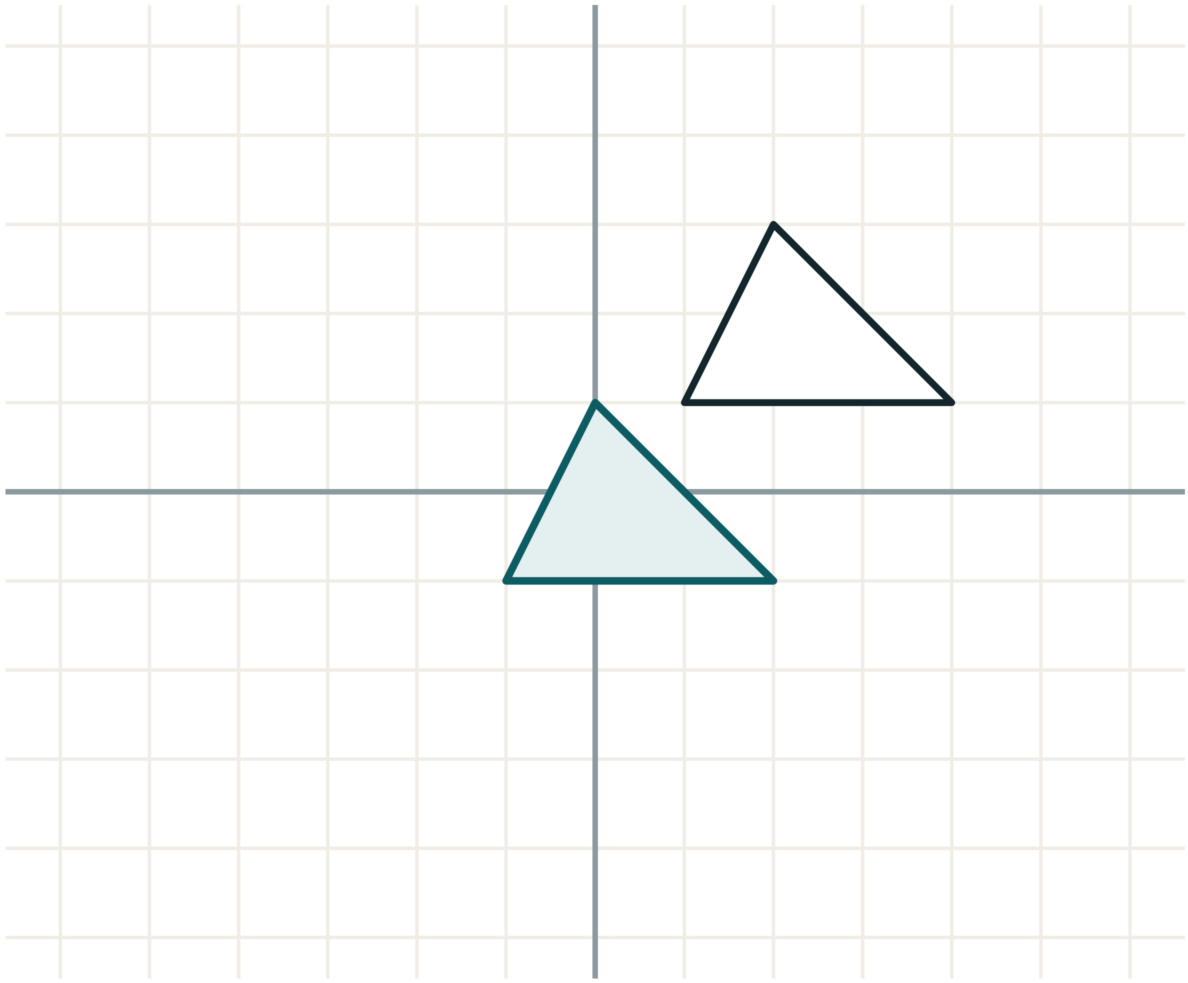
ANSWER

Order 6, 6 axes, and a centre

03 Interactive model

Pin a cut-out shape at a point and turn it; the class counts how many times it looks the same in one full turn, and that count is the order.

Turn about a point



transformation `translation` | lengths `unchanged` | angles `unchanged` | orientation `kept`

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Central symmetry	Markazga nisbatan simmetriya	Центральная симметрия
Centre of symmetry	Simmetriya markazi	Центр симметрии
Rotation	Burish	Поворот
Angle of rotation	Burilish burchagi	Угол поворота
Direction	Yo‘nalish	Направление
Half-turn	Yarim burilish	Поворот на 180°
Rotational symmetry	Burilish simmetriyasi	Поворотная симметрия
Order of symmetry	Simmetriya tartibi	Порядок симметрии

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Central symmetry is a rotation of:

- $90°$
- $180°$
- $270°$
- $360°$

(x, y) under a half-turn about O:

- $(-x, y)$
- $(x, -y)$
- $(-x, -y)$
- (y, x)

$90°$ anticlockwise sends $(3, 1)$ to:

- $(1, 3)$
- $(-1, 3)$
- $(1, -3)$
- $(-3, -1)$

A rotation is described by:

- a vector
- an axis
- centre, angle and direction
- a scale factor

A regular pentagon has a centre of symmetry:

- yes
- no
- sometimes
- five of them

A parallelogram has rotational symmetry of order:

- 1
- 2
- 3
- 4

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $(3, 1)$ under $180°$

   ANSWER $(-3, -1)$

2. $(3, 1)$ under $90°$ anticlockwise

   ANSWER $(-1, 3)$

3. $(3, 1)$ under $270°$ anticlockwise

   ANSWER $(1, -3)$

4. $(-2, 5)$ under $180°$

   ANSWER $(2, -5)$

5. Order of a square

   ANSWER 4

6. Order of an equilateral triangle

   ANSWER 3

7. Centre of symmetry of a parallelogram

   ANSWER Where the diagonals meet

Medium · the standard question

7 PROBLEMS

1. $(4, -2) \to (-4, 2)$: the transformation

   ANSWER Half-turn about O

2. Order and axes of a regular hexagon

   ANSWER 6 and 6

3. Order and axes of a rectangle

   ANSWER 2 and 2

4. Has a regular pentagon a centre?

   ANSWER No

5. Has a regular hexagon a centre?

   ANSWER Yes

6. $90°$ clockwise is the same as

   ANSWER $270°$ anticlockwise

7. Smallest rotation for a regular octagon

   ANSWER $45°$

Hard · stretch and reason

7 PROBLEMS

1. The image of $y = 2x$ under a half-turn about O

   ANSWER $y = 2x$ — unchanged

2. The image of $y = x^2$ under a half-turn about O

   ANSWER $y = -x^2$

3. The image of $(1, 2)$ under $90°$ anticlockwise about $(1, 1)$

   ANSWER $(0, 1)$

4. Two half-turns about different centres give

   ANSWER A translation

5. A rotation of $120°$ three times gives

   ANSWER The identity

6. Which quadrilaterals have both an axis and a centre?

   ANSWER Rectangle, rhombus, square

7. Order of rotational symmetry of the letter S

   ANSWER 2

07 Homework

Homework — 5 tasks

Give centre, angle and direction in every description of a rotation.

1. Find the image of $(5, -2)$ under rotations of $90°$, $180°$ and $270°$ anticlockwise about the origin.
2. A point $(2, 7)$ maps to $(-2, -7)$. Describe the transformation fully.
3. Give the order of rotational symmetry and the number of axes for a regular decagon.
4. Which of the special quadrilaterals have a centre of symmetry?
5. Find the image of $(3, 4)$ under a half-turn about $(1, 1)$.

Similarity of geometric figures

The definition widened from polygons to any figure at all — including curved ones.

LESSON 13 | 1 × 40 MIN | GEOMETRIYA 9, §11 | IGX 11.3

LESSON CLOCK

10' | 12' | 12' | 6'

The general definition	10 min
Which figures are always similar	12 min
Lengths, areas, volumes	12 min
Homework	6 min

LEARNING OBJECTIVES

- Define similarity of arbitrary figures by a distance-scaling map.
- Recognise that all circles, all squares and all regular n-gons are similar.
- Use the k, k^2, k^3 rule for lengths, areas and volumes.
- Apply similarity to scale models and maps.

TEXTBOOK

National	pp. 57–60
Cambridge	Core & Extended, pp. 232–236
Workbook	Exercise 11.3

01 Explanation

The general definition

A **similarity transformation** with coefficient $k > 0$ is a map of the plane such that for every pair of points,

$$P'Q' = k \cdot PQ$$

Two figures are **similar** if some similarity transformation carries one onto the other. For polygons this reduces to the definition of the second lesson; but the new definition also applies to circles, arcs, sectors and any curved figure at all.

$K = 1$ GIVES THE MOVEMENTS

A similarity with $k = 1$ preserves every distance — it is a movement. So congruence is the special case of similarity in which nothing is resized, and the previous three lessons were about that case.

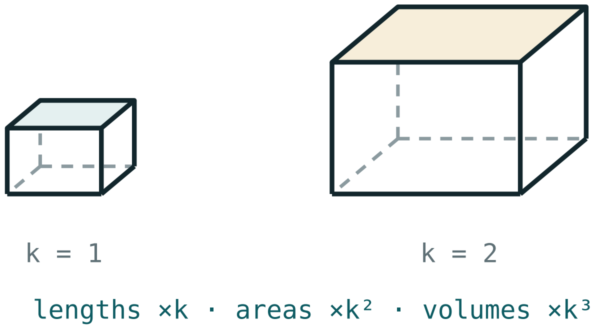
Which figures are always similar

PAIR	ALWAYS SIMILAR?	WHY
any two circles	yes	a circle is determined by its radius alone
any two squares	yes	determined by the side
any two regular n-gons (same n)	yes	angles fixed, sides equal
any two equilateral triangles	yes	the case $n = 3$
any two rectangles	**no**	the ratio of the sides can differ
any two isosceles triangles	**no**	the apex angle can differ
any two sectors of the same angle	yes	the shape depends only on the angle

> “SAME SHAPE” MUST MEAN “ONE SHAPE PARAMETER”
>
> A circle has no shape parameter at all, a regular n-gon none once n is fixed, but a rectangle has one — the ratio of its sides — and that is exactly why two rectangles need not be similar.

Lengths, areas, volumes

QUANTITY	MULTIPLIED BY	EXAMPLE, $K = 3$
length	k	$\times 3$
area	k^2	$\times 9$
volume	k^3	$\times 27$



Doubling every length multiplies the surface by four and the volume by eight.

Example. A model car is built at scale $1 : 20$. Its surface needs $\frac{1}{400}$ of the paint and it holds $\frac{1}{8000}$ of the volume — which is why small models are so much lighter than they look.

GOING BACKWARDS NEEDS A ROOT

Given the ratio of areas, k is the **square root**; given the ratio of volumes, the **cube root**. Areas $9 : 25$ give $k = \frac{3}{5}$; volumes $8 : 27$ give $k = \frac{2}{3}$.

02 Worked examples

EXAMPLE 1 **Two similar solids have volumes 54 and 128. Find the ratio of their surface areas.**

1. $k^3 = \dfrac{128}{54} = \dfrac{64}{27}$
2. $k = \dfrac{4}{3}$

   Cube root.
3. Areas scale by k^2.
4. $k^2 = \dfrac{16}{9}$, so $16 : 9$.

ANSWER

$16 : 9$

EXAMPLE 2 **A map has scale $1 : 25\ 000$. A lake covers $8\ cm^2$ on the map. Find its real area in km².**

1. $k = 25\ 000$ for lengths.
2. Areas scale by $k^2 = 6.25 \times 10^8$.
3. $8 \times 6.25 \times 10^8 = 5 \times 10^9\ cm^2$
4. $= 5 \times 10^5\ m^2 = 0.5\ km^2$

ANSWER

$0.5\ km^2$

EXAMPLE 3 **Are all rectangles similar? Justify with an example.**

1. All angles are $90°$ in every rectangle.
2. But a 1×2 and a 1×3 rectangle:
3. $\frac{1}{1} = 1$ while $\frac{2}{3}$ — not equal.
4. No. Their side ratios differ.

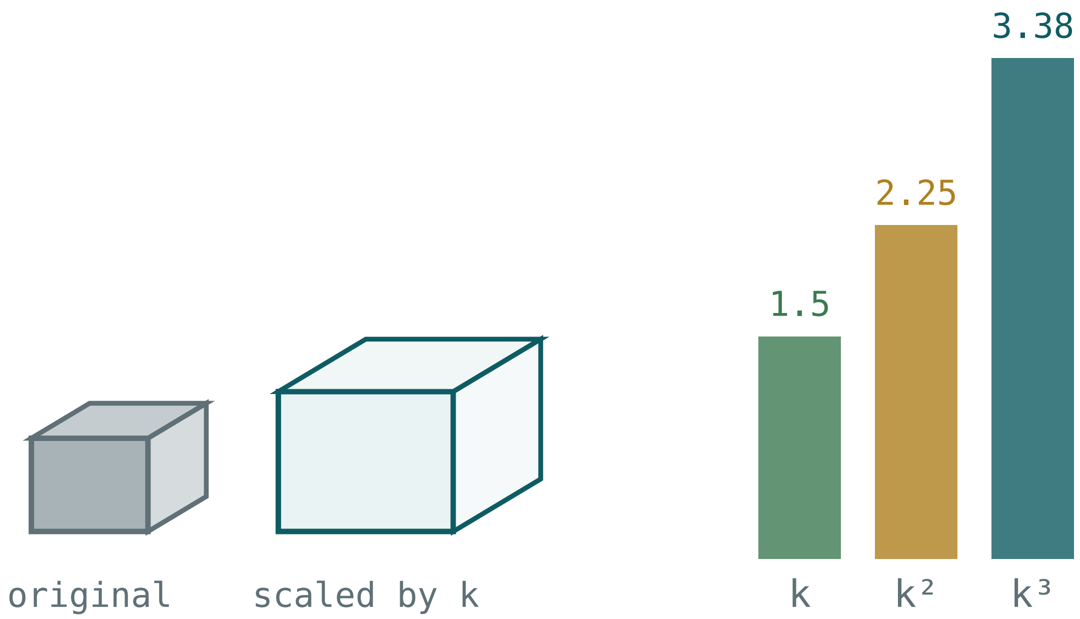
ANSWER

No

03 Interactive model

Show a matchbox and a shipping container photograph at the same apparent size; ask what would happen to weight if one really were a scaled copy of the other.

k, k² and k³



k **1.5** | length × **1.5** | area ×k² **2.25** | volume ×k³ **3.38**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Similarity transformation	O‘xshashlik almashtirishi	Преобразование подобия
Figure	Shakl	Фигура
Scale factor	Masshtab koeffitsiyenti	Масштабный коэффициент
Volume	Hajm	Объём
Model	Model	Модель
Map scale	Xarita masshtabi	Масштаб карты
Enlargement	Kattalashtirish	Увеличение
Reduction	Kichraytirish	Уменьшение

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A similarity multiplies every distance by:

- k
- k^2
- k^3
- 1

$k = 1$ gives:

- an enlargement
- a movement
- a reduction
- nothing

Any two circles are:

- congruent
- similar
- equal
- unrelated

Any two rectangles are:

- similar
- not necessarily similar
- congruent
- equal in area

Volumes scale by:

- k
- k^2
- k^3
- $3k$

Volumes $8 : 27$ give k =

- $\frac{8}{27}$
- $\frac{2}{3}$
- $\frac{4}{9}$
- $\frac{3}{2}$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $k = 2$: ratio of areas

   ANSWER 4

2. $k = 2$: ratio of volumes

   ANSWER 8

3. $k = 3$: ratio of volumes

   ANSWER 27

4. Areas $9 : 25$: k

   ANSWER $3 : 5$

5. Volumes $8 : 27$: k

   ANSWER $2 : 3$

6. Are all circles similar?

   ANSWER Yes

7. Are all rectangles similar?

   ANSWER No

Medium · the standard question

7 PROBLEMS

1. Volumes 54 and 128: ratio of areas

   ANSWER $16 : 9$

2. A model at $1 : 20$: ratio of surface areas

   ANSWER $1 : 400$

3. Same: ratio of volumes

   ANSWER $1 : 8000$

4. Map $1 : 25\,000$: $8\ cm^2$ in km²

   ANSWER 0.5

5. Areas 12 and 27: k

   ANSWER $2 : 3$

6. Are all regular hexagons similar?

   ANSWER Yes

7. Are all isosceles triangles similar?

   ANSWER No

Hard · stretch and reason

7 PROBLEMS

1. Two similar cones: heights 6 and 9, smaller volume 24π

   ANSWER 81π

2. Two similar cylinders: volumes 16 and 54, smaller radius 2

   ANSWER 3

3. A statue $2\ m$ tall weighs $160\ kg$; a model $0.5\ m$ tall

   ANSWER $2.5\ kg$

4. A map $1 : 50\,000$: a field of $0.25\ km^2$ on the map

   ANSWER $1\ cm^2$

5. Two similar solids: surface areas 50 and 72, ratio of volumes

   ANSWER $125 : 216$

6. A cone is cut halfway up: the ratio of the small cone to the whole

   ANSWER $1 : 8$

7. The ratio of the frustum to the whole cone

   ANSWER $7 : 8$

07 Homework

Homework — 5 tasks

Say which of k, k^2 or k^3 each question needs before computing.

1. Two similar solids have volumes 27 and 125. Find the ratio of their surface areas.
2. A model aeroplane is built at $1 : 72$. Find the ratio of the wing areas.
3. A map has scale $1 : 10\,000$. A park covers $5\ cm^2$. Find its real area in hectares.
4. Explain why all regular pentagons are similar but not all isosceles triangles are.
5. A cone is cut by a plane halfway up. Find the ratio of the volumes of the two pieces.

Properties of similar polygons

The consequences of similarity, gathered — and the area theorem proved rather than asserted.

LESSON 14 | 1 × 40 MIN | GEOMETRIYA 9, §12 | IGX 11.3

LESSON CLOCK

10' | 14' | 10' | 6'

The list of properties	10 min
Proving the area theorem	14 min
Backwards problems	10 min
Homework	6 min

LEARNING OBJECTIVES

- Prove that the ratio of the areas of similar polygons is k^2.
- Use the ratio of perimeters, diagonals and any corresponding lengths.
- Solve problems in which the area ratio is given and k is required.
- Apply the properties to a composite figure.

TEXTBOOK

National	pp. 61–65
Cambridge	Core & Extended, pp. 232–236
Workbook	Exercise 11.3

01 Explanation

The list of properties

Let two polygons be similar with coefficient k. Then:

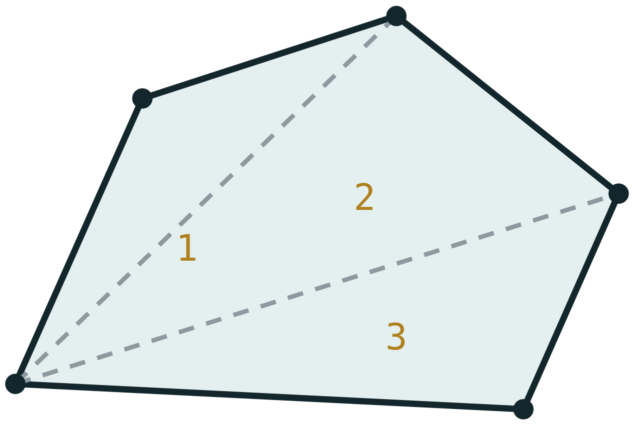
QUANTITY	RATIO
corresponding sides	k
corresponding diagonals	k
corresponding heights and medians	k
perimeters	k
radii of inscribed and circumscribed circles	k
corresponding angles	1
areas	k^2

ONE RULE, NOT SEVEN

Every **length** in the figure is multiplied by k; every **angle** is unchanged; every **area** is multiplied by k^2. The table is a list of examples of those three statements, not seven separate facts to learn.

Proving the area theorem

For triangles the proof is immediate: $S = \frac{1}{2}ah$, and both the base and the height are multiplied by k, so $S' = \frac{1}{2}(ka)(kh) = k^2S$.



Any polygon splits into triangles; each one scales by k^2, so the whole does.

For a general polygon, cut it into triangles by diagonals from one vertex. The similar polygon splits into corresponding triangles, each similar with the same k. Adding:

$$S' = k^2S_1 + k^2S_2 + \ldots = k^2(S_1 + S_2 + \ldots) = k^2S$$

DECOMPOSITION IS THE STANDARD METHOD FOR AREA PROOFS

Cut into triangles, prove it for a triangle, add up. The same argument gives the area of a trapezium, the area of a regular polygon, and — in the limit — the area of a disc in Quarter III.

Backwards problems

The examination favourite: the ratio of areas is given, and something linear is required.

GIVEN	STEP	THEN
areas $S_1 : S_2$	$k = \sqrt{S_1 : S_2}$	any length scales by k
perimeters $P_1 : P_2$	$k = P_1 : P_2$	areas scale by k^2

Example. Two similar polygons have areas 48 and 75. Then $k^2 = \frac{75}{48} = \frac{25}{16}$, so $k = \frac{5}{4}$ and the perimeters are in the ratio $5 : 4$.

DO NOT GIVE THE RATIO OF AREAS WHEN A LENGTH WAS ASKED FOR

$\frac{75}{48}$ is the ratio of the areas; $\frac{5}{4}$ is the ratio of the sides. Reading the question for which of the two is wanted is worth as much as the arithmetic.

02 Worked examples

EXAMPLE 1 **Two similar polygons have areas *48* and *75*. Find the ratio of their perimeters.**

1. $k^2 = \frac{75}{48}$
2. Cancel by 3: $\frac{25}{16}$.
3. $k = \frac{5}{4}$

   Square root.
4. Perimeters $5 : 4$.

ANSWER

$5 : 4$

EXAMPLE 2

A polygon of area *20* and perimeter *18* is enlarged so that its perimeter becomes *27*. Find its new area.

1. $k = \frac{27}{18} = 1.5$
2. Areas scale by $k^2 = 2.25$.
3. 20×2.25
4. $= 45$

ANSWER

45

EXAMPLE 3

A quadrilateral has diagonals *10* and *14* and area *56*. A similar quadrilateral has a diagonal of *21* corresponding to the *14*. Find its area.

1. $k = \frac{21}{14} = 1.5$

   Diagonals are lengths.
2. $k^2 = 2.25$
3. 56×2.25
4. $= 126$

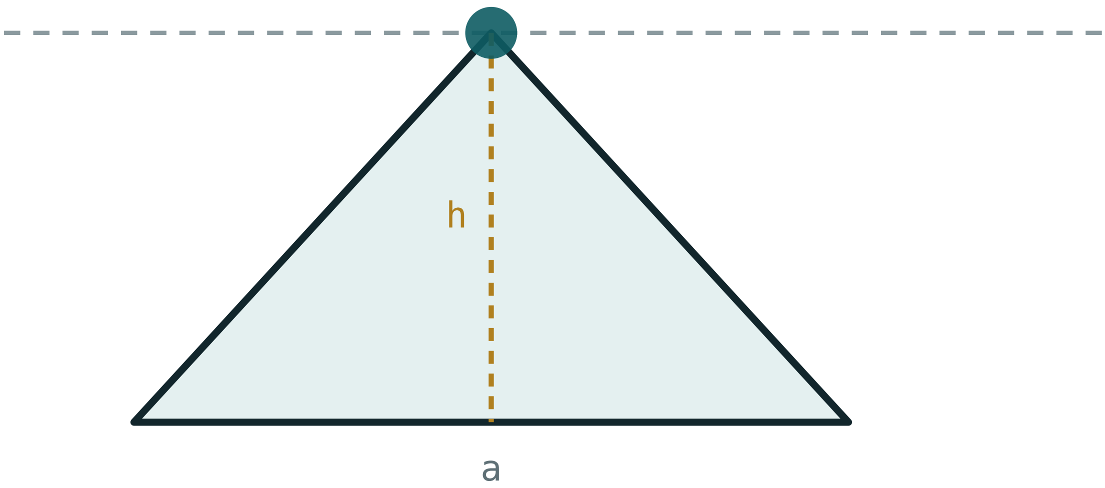
ANSWER

126

03 Interactive model

Draw a pentagon, cut it into triangles from one vertex, and scale each triangle on the board; the sum makes the k^2 proof visible in one picture.

Areas under enlargement



base a **220** | height h **120** | area ½ah **13200** | perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Property	Xossa	Свойство
Ratio of areas	Yuzalar nisbati	Отношение площадей
Ratio of perimeters	Perimetrlar nisbati	Отношение периметров
Diagonal	Diagonal	Диагональ
Decomposition	Bo‘laklarga ajratish	Разбиение
Corresponding	Mos	Соответственный
Composite figure	Murakkab shakl	Составная фигура
Theorem	Teorema	Теорема

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Corresponding diagonals are in the ratio:

- 1
- k
- k^2
- k^3

Areas are in the ratio:

- k
- k^2
- k^3
- $2k$

The proof for a polygon uses:

- Pythagoras
- decomposition into triangles
- the sine rule
- coordinates

Areas *48* and *75* give k =

$\frac{75}{48}$

$\frac{5}{4}$

$\frac{4}{5}$

$\frac{25}{16}$

Perimeter *18* → *27* scales the area by:

1.5

2.25

3

4.5

The inradius scales by:

1

k

k^2

it does not

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up
7 PROBLEMS

1. $k = 1.5$: ratio of areas

   ANSWER 2.25

2. $k = 4$: ratio of perimeters

   ANSWER 4

3. Areas $16 : 25$: k

   ANSWER $4 : 5$

4. Areas $4 : 9$: ratio of perimeters

   ANSWER $2 : 3$

5. Perimeters $3 : 7$: ratio of areas

   ANSWER $9 : 49$

6. Diagonals scale by

   ANSWER k

7. Angles scale by

   ANSWER 1

Medium · the standard question
7 PROBLEMS

1. Areas 48 and 75: ratio of perimeters

ANSWER $5 : 4$

2. Area 20, perimeter $18 \to 27$: the new area

ANSWER 45

3. Diagonal $14 \to 21$, area 56: the new area

ANSWER 126

4. Areas 50 and 98: k

ANSWER $5 : 7$

5. Perimeter 24, $k = \dfrac{5}{6}$: the new perimeter

ANSWER 20

6. Inradius 6, $k = 2.5$: the new inradius

ANSWER 15

7. Areas 27 and 12: k

ANSWER $3 : 2$

Hard · stretch and reason

7 PROBLEMS

1. Two similar polygons: areas differ by 44 and $k = \frac{5}{3}$

   ANSWER Areas 24.75 and 68.75

2. A polygon and its enlargement have total area 125 with $k = 2$

   ANSWER 25 and 100

3. Two similar triangles with perimeters 30 and 42, larger area 98

   ANSWER 50

4. A rectangle 6×8 is enlarged to area 192: k

   ANSWER 2

5. Its new diagonal

   ANSWER 20

6. Two similar polygons with equal areas: k

   ANSWER 1 — they are congruent

7. A figure is scaled so that the area triples: k

   ANSWER $\sqrt{3}$

07 Homework

Homework — 5 tasks

Say whether the quantity asked for is a length or an area before choosing k or k^2.

1. Two similar polygons have areas 27 and 48. Find the ratio of their perimeters.
2. A polygon of area 35 has its perimeter increased from 20 to 30. Find its new area.
3. Two similar triangles have perimeters 16 and 24. The smaller has area 30. Find the larger area.
4. Prove that the ratio of the areas of two similar polygons is k^2.

5. A figure is scaled so that its area is halved. Find k.

Homothety and similarity

The transformation that actually produces similar figures — a stretch from a fixed centre.

LESSON 15 | 1 × 40 MIN | GEOMETRIYA 9, §13 | IGX 11.3

LESSON CLOCK

10' | 14' | 10' | 6'

The definition	10 min
Constructing the image	14 min
Homothety and similarity	10 min
Homework	6 min

LEARNING OBJECTIVES

- Define homothety with centre O and coefficient k.
- Construct the image of a figure under a homothety, including negative k.
- Find images under a homothety centred at the origin.
- Know that every similarity is a homothety followed by a movement.

TEXTBOOK

National	pp. 66–71
Cambridge	Core & Extended, pp. 232–236
Workbook	Exercise 11.3

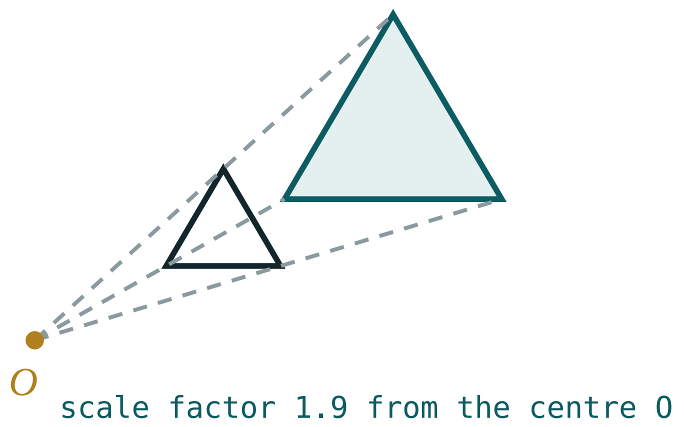
01 Explanation

The definition

A **homothety** with centre O and coefficient $k \neq 0$ sends each point P to the point P' on the line OP with

$$OP' = |k| \cdot OP$$

on the same side of O if $k > 0$, on the opposite side if $k < 0$.



Rays from the centre carry every point outwards (or inwards) by the same factor.

K	EFFECT
$k > 1$	enlargement, same side
$0 < k < 1$	reduction, same side
$k = 1$	the identity
$k = -1$	central symmetry in O
$k < 0$	resize and turn through $180°$

CAMBRIDGE CALLS IT AN ENLARGEMENT, AND ALLOWS NEGATIVE SCALE FACTORS

The row $k = -1$ is worth noticing: central symmetry, met two lessons ago, is a homothety. The transformations of this chapter form one family, not five unrelated rules.

Constructing the image

With the centre at the origin the formula is as simple as it can be:

$$P(x, y) \mapsto P'(kx, ky)$$

POINT	$K = 2$	$K = \frac{1}{2}$	$K = -1$
$(3, 4)$	$(6, 8)$	$(1.5, 2)$	$(-3, -4)$
$(-2, 6)$	$(-4, 12)$	$(-1, 3)$	$(2, -6)$

With a centre $C(a, b)$ other than the origin, work relative to C:

$$P' = (a + k(x - a), b + k(y - b))$$

To construct without coordinates: draw the ray from O through each vertex, and mark the new point at k times the distance. Joining the marks gives the image.

THE CENTRE IS THE ONLY POINT THAT DOES NOT MOVE

Every other point slides along its ray. A construction in which O has moved is wrong, and that single check catches most errors.

Homothety and similarity

A homothety with coefficient k is a similarity transformation with coefficient |k|

So a figure and its homothetic image are always similar. The converse is almost true and is the structural fact of the chapter:

Every similarity is a homothety followed by a movement

TRANSFORMATION	PRESERVES DISTANCE?	PRESERVES SHAPE?
translation, reflection, rotation	yes	yes
homothety	no	yes
a general similarity	no	yes

THE WHOLE CHAPTER IN ONE SENTENCE

Movements are similarities with $k = 1$; homotheties supply every other k; and combining the two produces every similarity there is. Nothing else is needed.

02 Worked examples

EXAMPLE 1 **Find the image of $A(3, 4)$ under a homothety centred at O with $k = 2$, and with $k = -1$.**

1. $k = 2$: multiply both coordinates.
2. $(6, 8)$
3. $k = -1$: $(-3, -4)$.
4. The second is central symmetry in O.

ANSWER

$(6, 8)$ and $(-3, -4)$

EXAMPLE 2 **A homothety with centre $C(1, 2)$ and $k = 3$ maps $P(4, 6)$. Find its image.**

1. Relative to C: $(4 - 1, 6 - 2) = (3, 4)$.
2. Multiply by 3: $(9, 12)$.
3. Add C back: $(1 + 9, 2 + 12)$.
4. $P'(10, 14)$

ANSWER

$(10, 14)$

EXAMPLE 3 **A triangle of area 12 undergoes a homothety with $k = -2$. Find the area of the image.**

1. Lengths scale by $|k| = 2$.
   The sign only turns the figure.
2. Areas scale by $|k|^2 = 4$.
3. 12×4
4. $= 48$

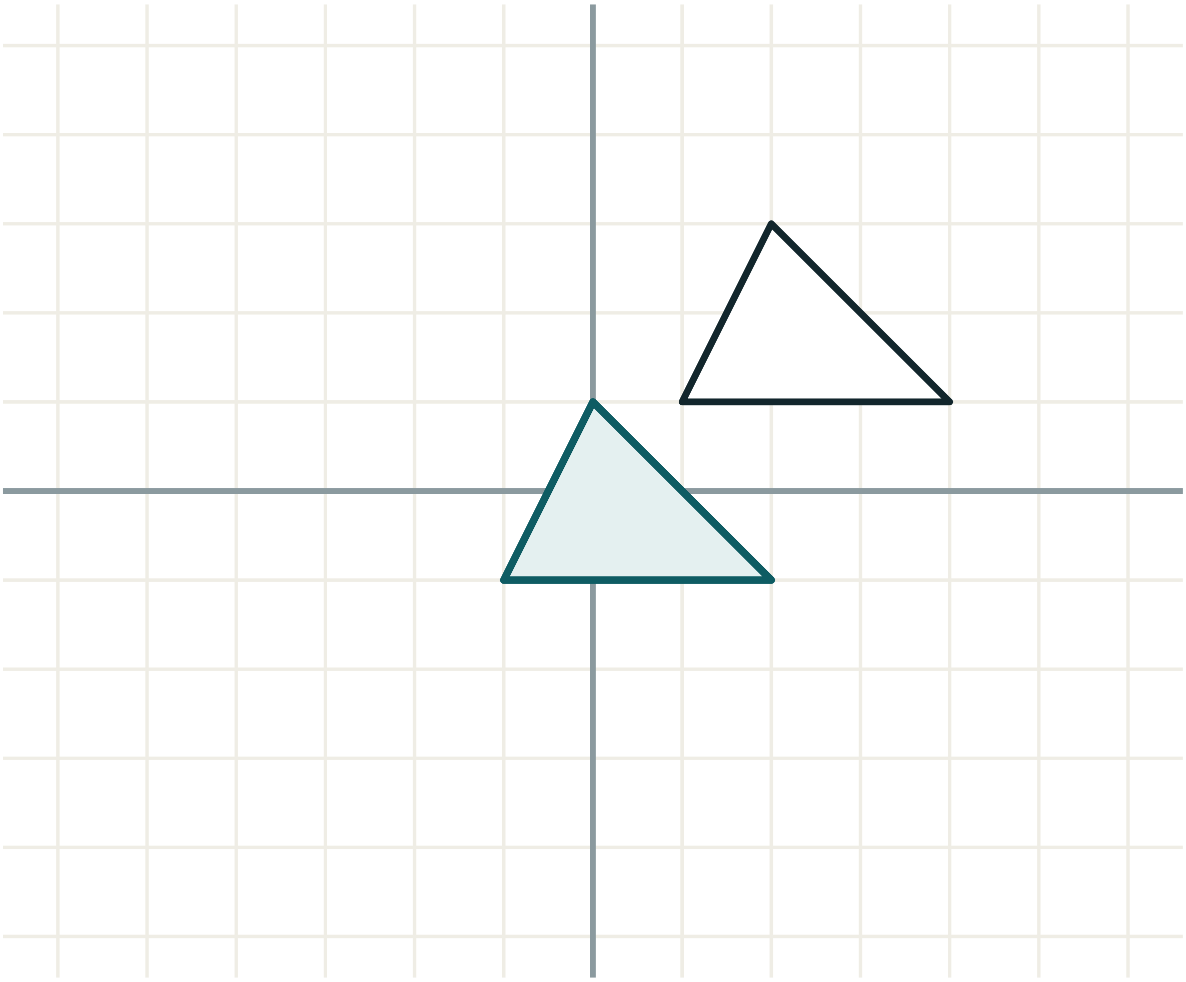
ANSWER

48

03 Interactive model

Use a torch and a cut-out shape to cast a shadow on the wall; the bulb is the centre, and moving it changes k in front of the class.

Stretch from a centre



transformation `translation` | lengths `unchanged` | angles `unchanged` | orientation `kept`

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Homothety	Gomotetiya	Гомотетия
Centre of homothety	Gomotetiya markazi	Центр гомотетии
Coefficient	Koeffitsiyent	Коэффициент
Enlargement	Kattalashtirish	Увеличение
Negative coefficient	Manfiy koeffitsiyent	Отрицательный коэффициент
Ray	Nur	Луч
Corresponding points	Mos nuqtalar	Соответственные точки
Invariant point	O‘zgarmas nuqta	Инвариантная точка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

A homothety is described by:

- a vector
- a centre and a coefficient
- an axis
- an angle

$k = -1$ gives:

- the identity
- a translation
- central symmetry
- a reflection

$(3, 4)$ under $k = 2$ about O:

- $(5, 6)$
- $(6, 8)$
- $(1.5, 2)$
- $(-3, -4)$

The only fixed point is:

- the origin
- the centre
- every point
- none

Areas scale by:

- k
- $|k|$
- k^2
- $|k|^3$

Every similarity is:

- a homothety
- a movement
- a homothety followed by a movement
- a reflection

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $(3, 4)$, $k = 2$ about O

   ANSWER $(6, 8)$

2. $(3, 4)$, $k = \frac{1}{2}$ about O

   ANSWER $(1.5, 2)$

3. $(3, 4)$, $k = -1$ about O

   ANSWER $(-3, -4)$

4. $(-2, 6)$, $k = 2$ about O

   ANSWER $(-4, 12)$

5. $k = 1$ gives

   ANSWER The identity

6. Fixed point of a homothety

   ANSWER The centre

7. Areas scale by

   ANSWER k^2

Medium · the standard question

7 PROBLEMS

1. $P(4, 6)$, centre $(1, 2)$, $k = 3$

   ANSWER $(10, 14)$

2. $P(5, 1)$, centre $(1, 1)$, $k = 2$

   ANSWER $(9, 1)$

3. Area 12, $k = -2$: the image area

   ANSWER 48

4. Area 50, $k = \frac{1}{5}$: the image area

   ANSWER 2

5. $k = -1$ is the same as

   ANSWER Central symmetry

6. A homothety with $k = 3$ applied twice

   ANSWER $k = 9$

7. Perimeter 30, $k = -1.5$

   ANSWER 45

Hard · stretch and reason

7 PROBLEMS

1. The image of $y = 2x + 4$ under $k = 2$ about O

   ANSWER $y = 2x + 8$

2. The image of $x^2 + y^2 = 4$ under $k = 3$ about O

   ANSWER $x^2 + y^2 = 36$

3. A homothety maps $(2, 3)$ to $(6, 9)$: find k if the centre is O

   ANSWER 3

4. A homothety centred at $(2, 2)$ maps $(4, 6)$ to $(8, 14)$: find k

   ANSWER 3

5. Composing homotheties with k_1 and k_2 about the same centre

   ANSWER $k_1 k_2$

6. A homothety with $k = -2$ followed by one with $k = -\frac{1}{2}$

   ANSWER The identity

7. Why is a homothety not a movement?

   ANSWER It changes distances unless $|k| = 1$

07 Homework

Homework — 5 tasks

Draw the rays from the centre in every construction; the image is read off them.

1. Find the image of $(5, -3)$ under homotheties about O with $k = 3$, $k = \frac{1}{2}$ and $k = -2$.
2. A homothety with centre $(2, 1)$ and $k = 2$ maps $(5, 4)$. Find its image.
3. A polygon of area 18 undergoes a homothety with $k = -3$. Find the image area.
4. Explain why central symmetry is a homothety.

5. Find the image of the circle $x^2 + y^2 = 9$ under a homothety about O with $k = 2$.

Chapter exercises — Cambridge congruence and similarity notation

The same theorems in the language and the question style of an IGCSE paper.

LESSON 16 | 1 × 40 MIN | GEOMETRIYA 9, I BOB MASHQLARI | IGX 11.4

LESSON CLOCK

10' | 12' | 12' | 6'

The four congruence conditions	10 min
Describing a transformation fully	12 min
Exam-style items	12 min
Homework	6 min

LEARNING OBJECTIVES

- Use the Cambridge names for the congruence conditions: SSS, SAS, ASA, RHS.
- Describe a transformation fully in the form an IGCSE paper requires.
- Answer “prove that these triangles are congruent” with reasons in the expected form.
- Use scale-factor language for enlargement, including a negative scale factor.

TEXTBOOK

National	pp. 72–75
Cambridge	Core & Extended, pp. 237–241
Workbook	Exercise 11.4

01 Explanation

The four congruence conditions

CODE	MEANS	UZBEK CRITERION
SSS	three sides	the third
SAS	two sides and the included angle	the first
ASA	two angles and the side between	the second
AAS	two angles and a side not between	a consequence of ASA
RHS	right angle, hypotenuse and one side	the right-triangle criterion

AAA IS NOT A CONGRUENCE CONDITION

Three equal angles give **similarity**, never congruence — the triangles can be any size. This is the single most common wrong answer in IGCSE congruence questions.

Describing a transformation fully

TRANSFORMATION	MUST STATE	EXAMPLE OF A FULL ANSWER
translation	the vector	a translation by $(3, -2)$
reflection	the mirror line	a reflection in $y = x$
rotation	centre, angle, direction	a rotation of $90°$ clockwise about $(0, 0)$
enlargement	centre and scale factor	an enlargement, centre $(1, 1)$, scale factor 2

ONE TRANSFORMATION ONLY

If a question says "describe the **single** transformation", an answer of the form "a reflection and then a translation" scores zero, even when it is geometrically correct. Find the one transformation that does the whole job.

Exam-style items

QUESTION	WHAT EARNS THE MARKS
"Prove $\triangle ABC \equiv \triangle DEF$"	three statements with reasons, then the condition named
"Describe fully the single transformation"	the type plus its data
"Find the scale factor"	a ratio of corresponding lengths, with the direction stated
"Explain why the triangles are similar"	two equal angles, each with a reason
"Work out the area of the larger shape"	k^2

A worked model answer for the first row:

STATEMENT	REASON
$AB = DE$	given
$\angle ABC = \angle DEF$	given
$BC = EF$	given
$\triangle ABC \equiv \triangle DEF$	*SAS*

02 Worked examples

EXAMPLE 1 **A shape at $(1,1)$, $(3,1)$, $(1,4)$ maps to $(2,2)$, $(6,2)$, $(2,8)$. Describe the single transformation fully.**

1. Each coordinate is doubled.
2. That is an enlargement about the origin.
3. Scale factor 2.
4. An enlargement, centre $(0, 0)$, scale factor 2.

   Type plus data.

ANSWER

Enlargement, centre $(0,0)$, scale factor 2

EXAMPLE 2 **Two triangles have a right angle, equal hypotenuses of 13 and one leg of 5 each. Name the condition.**

1. A right angle in each.
2. Equal hypotenuses.
3. One pair of equal sides.
4. RHS — congruent.

ANSWER

RHS

EXAMPLE 3 **A shape of area 7 is enlarged by scale factor 4. Find the new area.**

1. Lengths $\times 4$.
2. Areas $\times 4^2 = 16$.
3. 7×16
4. $= 112$

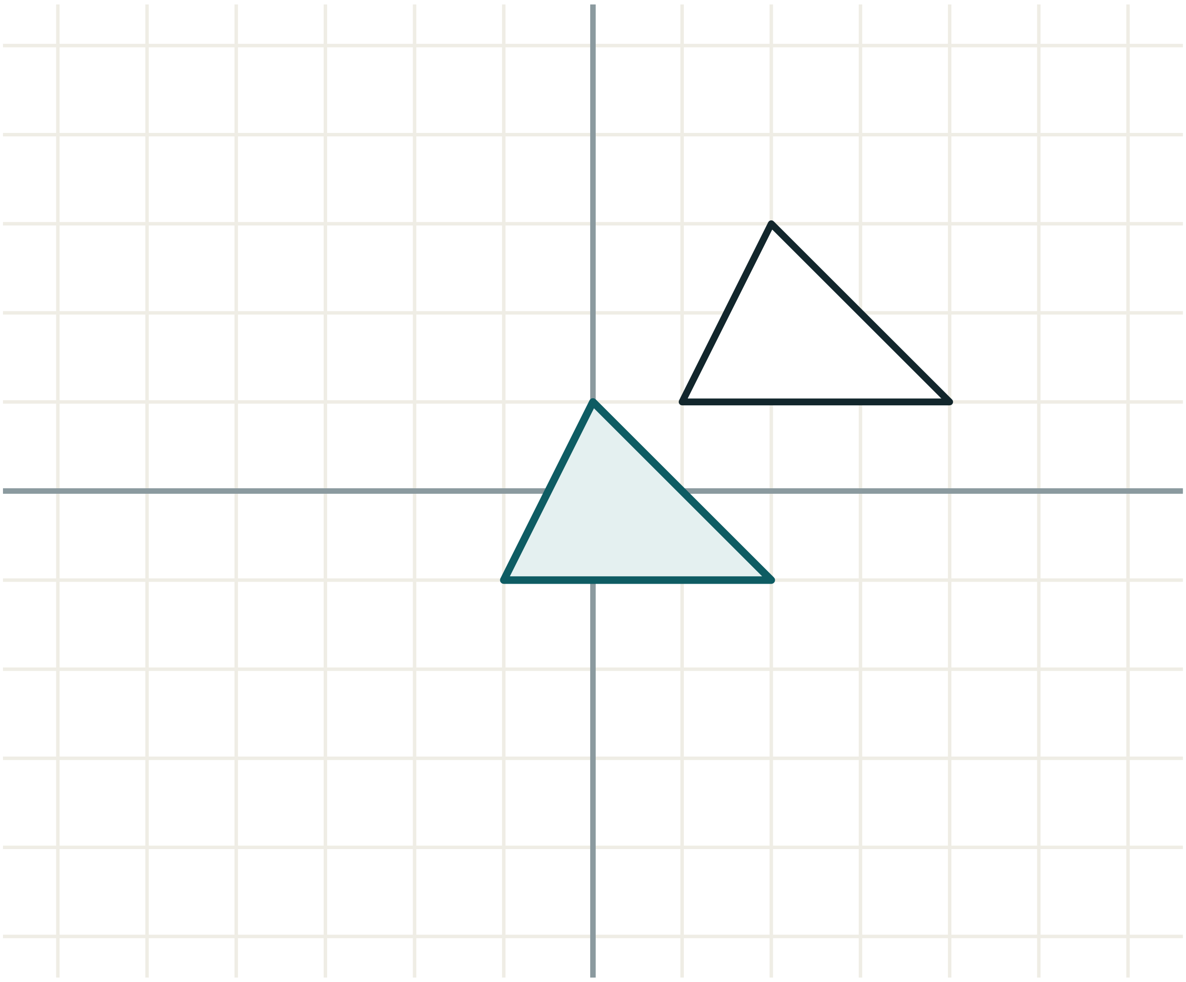
ANSWER

112

03 Interactive model

Read out four candidate answers to "describe fully" — one missing the centre, one missing the direction, one giving two transformations, one complete — and let the class mark them.

Name the transformation



transformation **translation** | lengths **unchanged** | angles **unchanged** | orientation **kept**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Congruent	Teng	Равный
SSS	USU	ССС
SAS	UBU	СУС
ASA	BUB	УСУ
RHS	TGK	ПГК
Enlargement	Kattalashtirish	Увеличение
Scale factor	Masshtab koeffitsiyenti	Масштабный коэффициент
Fully describe	To‘liq tasvirlash	Полностью описать

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

AAA gives:

- congruence
- similarity
- nothing
- equality of areas

RHS stands for:

- right, hypotenuse, side
- right, height, side
- ratio, height, side
- none

A rotation must state:

- the angle
- the centre
- the direction
- all three

An enlargement must state:

- the vector
- the centre and the scale factor
- the axis
- the angle

“Describe the single transformation” allows:

- two transformations
- one transformation
- any description
- a sketch only

Area 7, scale factor 4:

- 28
- 49
- 112
- 448

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Three equal sides

   ANSWER *SSS*

2. Two sides and the included angle

   ANSWER *SAS*

3. Two angles and the side between

   ANSWER *ASA*

4. Right angle, hypotenuse and a side

   ANSWER *RHS*

5. Three equal angles gives

   ANSWER Similarity only

6. A translation is described by

   ANSWER A vector

7. A reflection is described by

   ANSWER A mirror line

Medium · the standard question

7 PROBLEMS

1. $(1,1),(3,1),(1,4) \to (2,2),(6,2),(2,8)$

   ANSWER Enlargement, centre O, s.f. 2

2. $(1,2) \to (1,-2)$ with the shape flipped

   ANSWER Reflection in Ox

3. $(2,3) \to (-3,2)$

   ANSWER Rotation $90°$ anticlockwise about O

4. Area 7, scale factor 4

   ANSWER 112

5. Area 45, scale factor $\frac{1}{3}$

   ANSWER 5

6. A rotation description missing the centre scores

   ANSWER Fewer marks

7. Two triangles, hypotenuse 13 and leg 5

   ANSWER RHS

Hard · stretch and reason

7 PROBLEMS

1. An enlargement with s.f. -2 about O maps $(3, 1)$

   ANSWER $(-6, -2)$

2. An enlargement, centre $(1,1)$, s.f. 3, maps $(2, 3)$

   ANSWER $(4, 7)$

3. A shape and its image have areas 9 and 81: the scale factor

   ANSWER ± 3

4. Describe fully: $(x, y) \mapsto (y, x)$

   ANSWER Reflection in $y = x$

5. Describe fully: $(x, y) \mapsto (-x, -y)$

   ANSWER Rotation $180°$ about O

6. Describe fully: $(x, y) \mapsto (x + 2, y - 5)$

   ANSWER Translation by $(2, -5)$

7. Why is AAA not a congruence condition?

   ANSWER The triangles may differ in size

07 Homework

Homework — 5 tasks

Every description must give the type and all of its data.

1. Name the congruence condition for each of: three sides; two angles and the side between; a right angle with the hypotenuse and a leg.
2. Describe fully the transformation $(x, y) \mapsto (-y, x)$.
3. Describe fully the transformation $(x, y) \mapsto (3x, 3y)$.
4. A shape of area 12 is enlarged by scale factor 2.5. Find the new area.

5. Explain, with a diagram, why AAA does not prove congruence.

Control work 1, and work on the mistakes

Similarity and the transformations of the plane, tested — and the chapter closed as one map.

LESSON 17–18 | 2 × 40 MIN | GEOMETRIYA 9, NAZORAT ISHI 1 | IGX 11 REVIEW

LESSON CLOCK

3 | 40' | 12' | 20' | 5'

Instructions	3 min
The paper	40 min
Answers	12 min
Diagnosis and rewrite	20 min
The map	5 min

LEARNING OBJECTIVES

- Establish similarity by the appropriate criterion under time.
- Use k, k^2 correctly for lengths and areas.
- Describe a transformation fully.
- Classify each lost mark and rewrite the whole solution.

TEXTBOOK

National	pp. 9–75
Cambridge	Core & Extended, pp. 220–241
Workbook	Control paper G1

01 Explanation

The paper — 30 marks, 40 minutes

Q	TASK	MARKS	FROM
1	In $\triangle ABC$, $DE \parallel BC$, $AD = 5$, $DB = 3$, $DE = 10$: find BC	5	L5
2	Are triangles $4, 6, 8$ and $6, 9, 12$ similar? Name the criterion	5	L7
3	Two similar polygons have areas 27 and 48: find the ratio of their perimeters	5	L14
4	The altitude to the hypotenuse divides it into 4 and 16: find the altitude	5	L8
5	Reflect $(3, -2)$ in Ox, in Oy and in $y = x$	5	L11
6	Describe fully the transformation $(x, y) \mapsto (2x, 2y)$	5	L15

WHERE THE MARKS ACTUALLY GO

Q1 carries one mark for using $AB = 8$ rather than $DB = 3$; Q2 two for naming the criterion; Q3 two for taking the square root; Q5 one for the swap in $y = x$; Q6 two for giving the centre as well as the scale factor.

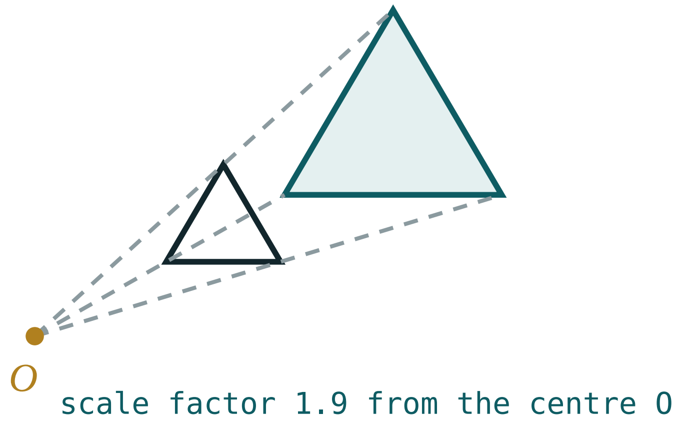
Naming the slip

SLIP	WHAT IT LOOKS LIKE	THE FIX
part of a side used as the whole	$k = \frac{3}{5}$	$k = \frac{8}{5}$
criterion not named	“they are similar”	“by the third criterion”
ratio of areas given for a length	$\frac{48}{27}$	$\frac{4}{3}$
wrong altitude relation	$CH^2 = 4 \times 20$	$CH^2 = 4 \times 16$
$y = x$	$(-3, 2)$	$(-2, 3)$
centre omitted	“an enlargement, s.f. 2”	“…centre $(0, 0)$”
sides not sorted	$\frac{6}{8} \neq \frac{9}{4}$	sort both lists first

Name the slip in the margin, then rewrite the whole solution — not the wrong line.

Chapter I as one map

BLOCK	THE SENTENCE
similarity of polygons	equal angles **and** proportional sides
similar triangles	either condition alone is enough
the three criteria	AA, SAS, SSS — proportionally
right triangles	the altitude makes three similar triangles
movements	translation, reflection, rotation — distance preserved
homothety	a stretch from a centre — shape preserved, size not
the ratios	lengths k, areas k^2, volumes k^3



One picture for the whole chapter — a figure and its scaled copy.

LOOKING FORWARD

Chapter II removes the right angle. The sine and cosine rules let a triangle be solved from any three suitable pieces of information, and the trigonometric ratios are extended to obtuse angles. Nothing from Chapter I is needed for the formulae, but the similar-triangle habits are used in every proof.

02 Worked examples

EXAMPLE 1 **Model answer, Q1: $DE \parallel BC$, $AD = 5$, $DB = 3$, $DE = 10$.**

1. $\angle A$ common; $\angle ADE = \angle ABC$ (corresponding).

   First criterion.

2. $AB = 5 + 3 = 8$

   The whole side.

3. $k = \dfrac{8}{5}$

4. $BC = 10 \times \dfrac{8}{5} = 16$

ANSWER

$BC = 16$

EXAMPLE 2 **Model answer, Q3: areas 27 and 48.**

1. $k^2 = \dfrac{48}{27} = \dfrac{16}{9}$

2. $k = \dfrac{4}{3}$

   Square root.

3. Perimeters scale by k.

4. $4 : 3$

ANSWER

$4 : 3$

EXAMPLE 3 **Model answer, Q6: describe $(x, y) \mapsto (2x, 2y)$.**

1. Every distance from the origin is doubled.
2. The origin itself does not move.
   It is the centre.
3. Shape preserved, size not — an enlargement.
4. An enlargement, centre $(0, 0)$, scale factor 2.
   Type plus data.

ANSWER

Enlargement, centre $(0,0)$, scale factor 2

03 Interactive model

Return Q6 with four candidate answers written on the board — one missing the centre, one naming two transformations — and let the class allocate the five marks.

Chapter I in twelve questions

Similar polygons need:

- equal angles only
- proportional sides only
- both
- neither

For triangles, equal angles give:

- nothing
- proportional sides
- equal sides
- equal areas

The first criterion needs:

- one angle
- two angles
- three sides
- two sides

The second criterion needs the angle to be:

- the largest
- included
- acute
- right

Six sides and no angles: use

- the first
- the second
- the third
- none

The altitude to the hypotenuse makes:

- two
- three
- four
- no

CH^2 equals:

- $AH \cdot AB$
- $AH \cdot HB$
- $HB \cdot AB$
- AB^2

A translation is described by:

- a vector
- a centre
- an axis
- an angle

A reflection reverses:

- distance
- angle size
- orientation
- area

Central symmetry is a rotation of:

- $90°$
- $180°$
- $270°$
- $360°$

A homothety is described by:

- a vector
- a centre and a coefficient
- an axis
- an angle

Areas scale by:

- k
- k^2
- k^3
- $2k$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Criterion	Alomat	Признак
Coefficient of similarity	O‘xshashlik koeffitsiyenti	Коэффициент подобия
Ratio of areas	Yuzalar nisbati	Отношение площадей
Transformation	Almashtirish	Преобразование
Homothety	Gomotetiya	Гомотетия
Full description	To‘liq tavsif	Полное описание
Diagnosis	Tashxis	Диагностика

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

In Q1 the ratio uses:

- DB
- AB
- DE
- AC

Q2 must include:

- a diagram
- the criterion
- a decimal
- an angle

Q3 needs:

- squaring
- a square root
- a cube root
- nothing

In Q4 the whole hypotenuse is:

- 4
- 16
- 20
- 12

Q5's $y = x$ image of $(3, -2)$:

- $(-3, 2)$
- $(-2, 3)$
- $(2, -3)$
- $(3, 2)$

Q6 must give:

- the type only
- the type and its data
- two transformations
- a sketch

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $DE \parallel BC$, $AD = 5$, $DB = 3$: AB

   ANSWER 8

2. Same: k

   ANSWER $\frac{8}{5}$

3. Same, $DE = 10$: BC

   ANSWER 16

4. $4, 6, 8$ and $6, 9, 12$

   ANSWER Similar, $k = 1.5$

5. Areas 27 and 48: k

   ANSWER $\frac{4}{3}$

6. Projections 4 and 16: the altitude

   ANSWER 8

7. $(3, -2)$ in Ox

   ANSWER $(3, 2)$

Medium · the standard question

7 PROBLEMS

1. $(3, -2)$ in Oy

ANSWER $(-3, -2)$

2. $(3, -2)$ in $y = x$

ANSWER $(-2, 3)$

3. Describe $(x, y) \mapsto (2x, 2y)$

ANSWER Enlargement, centre O, s.f. 2

4. Areas 27 and 48: ratio of perimeters

ANSWER $4 : 3$

5. Projections 4 and 16: the legs

ANSWER $4\sqrt{5}$ and $8\sqrt{5}$

6. Which criterion for $4, 6, 8$ and $6, 9, 12$?

ANSWER The third

7. A shape of area 9 with $k = 2$: its area

ANSWER 36

Hard · stretch and reason

7 PROBLEMS

1. $DE \parallel BC$, $[\triangle ADE] = 25$, $[\triangle ABC] = 64$: $AD : DB$

   ANSWER $5 : 3$

2. A right triangle with hypotenuse 20 and altitude 8: the projections

   ANSWER 4 and 16

3. Two similar polygons: perimeters 21 and 28, smaller area 45

   ANSWER 80

4. Describe $(x, y) \mapsto (-x, -y)$

   ANSWER Rotation $180°$ about O

5. Describe $(x, y) \mapsto (y, x)$

   ANSWER Reflection in $y = x$

6. A bisector from A with $AB = 12$, $AC = 18$, $BC = 20$: BD

   ANSWER 8

7. A triangle $6, 8, 10$: the altitude to the hypotenuse

   ANSWER 4.8

07 Homework

Homework — 5 tasks

Rewrite in full every question that lost a mark before Chapter II begins.

1. In $\triangle ABC$, $DE \parallel BC$, $AD = 6$, $DB = 2$, $DE = 9$. Find BC.
2. Two similar polygons have areas 50 and 72. Find the ratio of their perimeters.
3. The altitude to the hypotenuse divides it into 5 and 20. Find the altitude and the legs.
4. Reflect $(-4, 1)$ in Ox, Oy and $y = x$.
5. Describe fully the transformation $(x, y) \mapsto (-y, x)$.